
The Verification Gap: Auditing Proof Progress and Repair on IMO 2026*

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Abstract

In the IMO 2026 campaigns we examined, model graders decided which proofs counted as correct. We took 8 model-accepted proofs, none checked by an expert, and AI agents made 24 single edits: a false step a later line uses (fatal), an argument replaced by an announced omission (gap), or a false line nothing uses (harmless). GPT-5.6-Sol and Claude Opus 5 graded all 32 texts on a rubric that scores “a small error that is straightforwardly repairable” 5–6 of 7. Required of both, a pass mark of 5 accepted 13 of the 16 fatal or gap texts in each of two runs (8 of 11 without the five disputed labels; the other labels are unconfirmed), though a grader quoted the edited passage in each. Requiring 7 instead, or the prompt’s undefined “complete” field (both read post hoc), rejects every fatal edit in both runs but also most harmless ones. With a pass mark of 7, which grader seems more accurate flips in both runs on one choice: whether a harmless false line counts as a mistake. These are counts on hand-picked texts under one rubric. The study grew out of an audit of solving campaigns, where two of our campaign gates accepted a false monotonicity claim that the sequence 2, 1 refutes. We release edits, checks, prompts and responses.

1 Introduction

On IMO 2026 several frontier models were credited with all six problems by the campaign’s Claude-based verifier agents [Das, 2026], a year after gold-medal results on IMO 2025 [Google DeepMind, 2025, Wei, 2025, Huang and Yang, 2025]. The bottleneck was then not producing credited proofs but knowing that a proof is correct: the verification gap. We ask what an operator can establish from one grading response, without an expert regrade. Our answer, from a planted-defect study (§2): required of both graders, a pass mark of 5/7 accepted 13 of the 16 fatal-or-gap edits in each of two runs and rejected 0 of the 16 sound texts (originals and harmless edits), although a grader had quoted every accepted edit; the rubric places a straightforwardly repairable error at 5–6. Requiring 7 instead, or the prompt’s undefined “complete” field (both read post hoc), rejects every fatal edit but also most harmless ones. With a pass mark of 7, which grader seems more accurate flips in both runs on whether a false line nothing uses counts as a mistake (the truth convention, §4).

Position. Model judges inflate olympiad proof scores [Petrov et al., 2025a] and are flipped by trivial tokens [Zhao et al., 2025]; provers write convincing proofs of false statements [Petrov et al., 2025b]; generation, verification and repair have been trained jointly [Chen et al., 2026]. RefGrader [Mahdavi et al., 2025b] finds that models flag incorrect solutions reliably but miscalibrate partial credit; binary judging can approach human agreement [Dekoninck et al., 2025]; ProofBench [Ma et al., 2025] varies the grader’s reference context and instruction strictness; planted perturbations [Doddapaneni et al., 2024] and inserted bugs [McAleese et al., 2024] have tested evaluators; and Mahdavi et al. [2025a] classify proof fallacies. We propose no grader, run none of these workflows and have no expert grades. We add where the failure sits: GPT named every planted fatal error and

*This version corrects the one the reviewers read; Appendix Q summarizes the changes and points to the full list.

at least one grader quoted each, and the pass mark still accepted the texts (on announced deletions one grader often listed no error). In ProofBench’s terms our grade arm is *reference-free* (“None”: no solution material and an instruction not to recall one, which cannot remove what a model knows) and our witness and closure arms (§5–6) *reference-guided* (“MS”: the prompt carries a solution-derived marking scheme). The nearest in-paper analogue, the repair-tolerant witness-plus-reference gate (App. L.7), which checks the witness objections of §5 against a ledger from a published solution, accepted 2 of 16 broken texts (both disputed) and rejected 0 of 16 sound ones in one run, a zero its convention partly builds in (as written it rejects 5).

Contributions. In reading order: three accept rules on planted texts, with a worked example (Figure 1; §2–3); convention dependence (§4); a reference-guided comparator (§5); the campaign audit behind the design (§6).

2 The Planted-Defect Study

Bases. We took 8 proofs of four of the six problems (2 on P1, 3 on P3, 1 on P5, 2 on P6): 6 by GPT-5.6-Sol (4 from our campaigns, 2 public), 1 by Kimi K3 and 1 by Claude Fable 5. They were chosen among texts model evaluators had accepted, so they are hand-picked, not a sample; “accepted” means model verdicts, same-lab for five, and no expert checked any of them (routes: App. L.2).

Edits. One edit per class per base gives 24 single-edit texts, 32 with the originals. Classes follow a cite rule: *fatal*, a statement a later line cites is made false, which leaves the argument invalid as written, not the theorem false; *gap*, a load-bearing proof is replaced by a one-line assertion or brief sketch containing “(standard; details omitted)”, so every deletion announces itself (8 of 8; none of the other 24 texts has the phrase); *harmless*, a false line nothing later uses, usually a display the next line silently corrects. The study design is the author’s; AI agents implemented it: a coordinating Claude Fable 5.1 agent selected the bases, wrote the specification and suggested several edits, and three Claude Fable 5.1 subagents wrote every edit, its dependency argument and intended label, and the 16 executable checks; the author read no edit before the gates ran. A check certifies that the planted statement is false (for the P5 sign flip, only that its expansion is not an identity), not that a consumer fails; the fatal and harmless edits have one, the 8 deletions none. The edits were committed at launch, 20 s after the earliest request on an edited text, after 3 smoke-test screenings of one original, and never changed (412 of the 412 launch-day requests on edited texts carry the recorded hash).

Truth, graders and rules. *As written* (AW), any planted false line is an error (a natural false line in an original keeps the original’s label); *repair-tolerant* (RT), a false line nothing later cites is acceptable. The 16 *broken* texts are the fatal and gap edits; the 16 *sound* texts are the originals and harmless edits, sound under RT only. GPT-5.6-Sol and Claude Opus 5 graded every text with one reference-free prompt whose scale reads “7 = complete and correct proof; 5-6 = essentially complete, with minor gaps or a small error that is straightforwardly repairable”. The response also returns a boolean *complete*, which the prompt names but never defines. Both graders required, we read three rules from each response: the *pass mark* (grade ≥ 5), the *strict mark* (grade = 7) and the *completeness flag*. The flag is nested between the cuts at 6 and at 7: in both runs every response graded 7 sets it and none graded 5 or below does, and it departs from a 7 on 4 of all 64 responses in the first run and 2 in the re-query, each a GPT response graded 6. “Strict rules” means the strict mark, the flag and the witness rule of §5.

Protocol and disputes. The self-timestamped protocol prespecified four gates (pass mark, closure, witness, witness+reference) with their false accepts on broken texts, false rejects on originals and harmless acceptance (all four gates’ counts: App. L.7), the quote locus (quotes inside the edit) and a separate report for GPT-written bases. The rest is post hoc: the strict mark, the flag, the sweep, the fatal/gap split, the sound set, the convention comparison and routing, read from stored responses without new calls, and the same-day re-query of the grade calls, which no review requested (App. L.1). Figures are from the first run unless the re-query is named. Internal model-run reviews disputed five broken labels, never changed: all 4 P1 fatal and gap edits (each repairable in one line or omitting a standard step) and the P5 sign flip, whose displayed equality holds at every admissible instance (App. L.4). A base’s four texts share content, so we report counts, not rates.

Planted fatal edit (p6-gpt56-ours-F; one edit shown as two diff hunks):

– $Q = \{p : p \text{ is prime and } p \leq A\}$,

+ $Q = \{p : p \text{ is prime and } p \mid A\}$,

+ Only the primes dividing A can matter here, because every term shares a prime factor with $A = a_1$.

Executable check, in the proof’s notation ($P(x)$: the primes dividing x ; $\sigma(x) = P(x) \cap Q$; key lemma: σ of each term meets P of every term): with $A = 15$, $\sigma(18) = \{3\}$ misses $P(20) = \{2, 5\}$, so the key lemma is false; its conclusion $a_{n+T} = a_n + L$ with $L = 15$, $T = 4$ fails: $a_2 = 18$ but $a_{2+4} = 36 \neq a_2 + 15$.

Text	GPT-5.6-Sol		Claude Opus 5		both graders required		
	grade	complete	grade	complete	pass mark	strict mark	flag
original	7/7	T/T	7/7	T/T	✓/✓	✓/✓	✓/✓
fatal	6/6	F/F	5/6	F/F	✓/✓	×/×	×/×
gap	3/3	F/F	2/2	F/F	×/×	×/×	×/×
harmless	6/7	T/T	7/7	T/T	✓/✓	×/✓	✓/✓

Each cell: first run / same-day re-query. complete: T = true, F = false. Decisions: ✓ accept, × reject; pass mark = both graders at least 5, strict mark = both at 7, flag = both mark the text complete.

Figure 1: One base, four texts, two graders: p6-gpt56-ours, a GPT-5.6-Sol proof of P6 from our campaign. A later line uses the changed set Q (fatal by the cite rule); the check shows the edited proof’s own conclusion fails, and the proof’s own closing section refutes the inserted line. Both graders name the changed definition and its correction (all primes at most A), each response citing the text’s own statement of it, mark the text incomplete and grade it 5–6. One of 3 undisputed fatal texts that meet a post-hoc selection rule in the first run, chosen among them as App. L.7 explains; the witness arm’s P6 scheme names the prime bound, so it does not illustrate the witness rule.

3 A Pass Mark Inside the Repairable Band

Headline. Both graders required, the pass mark accepted 13 of the 16 broken texts (8 of 8 fatal, 5 of 8 gap) and rejected 0 of the 16 sound ones in both runs (Table 1); no decision at 5 moved in the re-query (0 of 64 responses). On all 13 accepted broken texts at least one grader quoted the edited passage in both runs (Claude on 13; on the disputed GPT-base P1 deletion the only quote is the announcement itself), and in the first run at least one marked the text incomplete on 12 (both on 8); of the 26 responses on them, 5 were a 7 (6 in the re-query). Without the disputed labels 8 of 11 broken texts passed in each run; in both runs a grader proposes a repair for every undisputed fatal edit too (App. L.4), so exclusion does not certify the rest. The 5–6 band explicitly admits a straightforwardly repairable error, so a pass at 5 does not certify a proof complete as written: the mismatch is between a partial-credit scale and its use as a full-proof acceptance gate.

What passed, by kind. Accepted fatal edits plant a false general lemma (3), a false graph inference (2), a pigeonhole miscount (1), a sign error (1) or a wrong definition (1); in both runs GPT named the planted error on 8 of 8 fatal edits and gave each a 6. On 4 of the 5 accepted gap edits GPT’s main-gap field named no gap, and in the first run its notes call 2 of them routine or adequately sketched (3 in the re-query). In both runs the refused gap edits are the P6 deletions and the deletion in the Claude-written P3 base, and no P1 or P5 deletion was refused: a post-hoc pattern on 8 texts, confounded with problem, with no mechanism claimed; 3 fatal edits re-plant natural failures of §6.

Sweep and the strict rules. Sweeping the cutoff (post hoc, first run) trades one error for the other: (broken accepted, sound rejected) is (13, 0) at ≥ 5 , (11, 0) at ≥ 6 and (1, 9) at $= 7$; no cutoff on this rubric keeps both low. The strict mark and the flag reject every fatal edit in both runs. In the first run the strict mark accepts 1 broken text and rejects 9 sound ones (8 harmless edits and the Kimi original, which carries an unused false remark), the flag 1 and 6 (all harmless): they differ on 3 sound texts on 3 bases, all in the flag’s favour. In the re-query the strict mark gives 1 and 8 and the flag 2 and 7, one text each way, so neither dominates. In both runs the broken texts they accept are disputed P1 deletions. The flag’s advantage over the pass mark is strictness, paid for on the harmless class.

GPT’s grade of 6 is a price, not blindness. In the first run GPT gave all 16 fatal and harmless edits a 6, 4 of them on the two bases it did not write (in the re-query 1 harmless edit got a 7), yet it quoted the harmless edit on 8 of 8 and graded it below its grade of the corresponding original on 7 of 8 bases (6 in the re-query); Claude gave 8 of 8 harmless edits a 7 (7 in the re-query). GPT’s pass

Table 1: What each rule does to each kind of text. Both graders must accept; the rules read one reference-free response. First run, re-query in parentheses where it differs; counts by recorded label (model-written, §2; five disputed, all among the pass mark’s 13 broken accepts), not rates.

Rule	fatal accepted (of 8)	gap accepted (of 8)	harmless rejected (of 8)	original rejected (of 8)	broken accepted (of 16)	sound rejected (of 16)
pass mark (≥ 5)	8	5	0	0	13	0
strict mark (= 7)	0	1	8 (7)	1	1	9 (8)
completeness flag	0	1 (2)	6 (7)	0	1 (2)	6 (7)

mark accepted 11 of 12 broken texts on GPT-written bases and 4 of 4 on the other two bases, too few to separate self-preference [Panickssery et al., 2024]. *Scope*: one rubric; a rubric that caps an unresolved load-bearing error below 5 was not run in this protocol.

4 Which Grader Looks More Accurate Depends on the Convention

Counting each grader’s errors against the intended labels over the 32 texts (post hoc), under each strict rule Claude is more accurate under the repair-tolerant convention and GPT under the as-written one. GPT/Claude errors under the strict mark are RT 12/2 and AW 4/10 in the first run, RT 11/4 and AW 5/10 in the re-query; under the witness gate (run once in this protocol) RT 5/1 and AW 3/7. Adding the flag (first run RT 10/2, AW 6/10; App. L.7), the better grader reverses under 3 of 3 strict rules in the first run and 2 of 2 in the re-query; under the pass mark Claude is ahead under both conventions. Only the 8 harmless edits change label, and in the first run the strict rules agree pairwise on 27–29 of 32 texts: one dependence seen three correlated ways. Relabelling the Kimi original as a reject, or setting the disputed labels aside, leaves it intact (App. L.7); Figure 1’s harmless row shows the mechanism. Which of these two graders is better is unidentified until the convention is stated.

5 Routing and a Reference-Guided Comparator

Disclosure. The witness and closure prompts carry the problem’s marking scheme (the steps a complete proof needs, the answer on P3 and P5, named wrong approaches), which the grade prompt forbids. The P1 scheme names as wrong the false lemma both P1 fatal edits plant, but the strict mark and the flag, whose graders never saw it, also reject those texts, so this collision bears only on the comparison with the pass mark. It also requires the Euclidean step both P1 gap edits delete, and the broken texts the strict mark and the flag accept are such deletions: the witness rule’s broken-text margin over them rests on this second collision. All 4 P1 broken texts are disputed, so setting them aside leaves no P1 text; the witness rule’s outcome cannot be credited to obligation checking alone.

Gate and routing. The witness rule asks each grader, per obligation of the scheme, for a status (written and valid, written but invalid, not written) with a verbatim quote and a witness, and accepts when both scored obligations are written and valid (App. B, J, L.1). As a gate (one run, both graders required), the witness rule accepted 0 of 16 broken texts and rejected 5 of 16 sound ones, all harmless edits, so 0 of the 8 originals (the prespecified count), and its gain over the strict mark is 1 disputed announced deletion. As a router (post hoc) that accepts or rejects on agreement and refers disagreements to a person, it referred 4 of 32 texts, passed 0 of 16 broken and refused 1 of 16 sound ones (one run); the refused text, a harmless edit copying a display from our campaign’s Nemotron P5 text, is false as written. In the first run the flag referred 10, passed 1 broken and refused 0 sound texts (9, 2 and 1 in the re-query), and the pass mark 2, 13 and 0, identical in the re-query. The witness rule’s referrals were all harmless edits; the flag’s were 6 harmless edits, 3 announced deletions and 1 sign error (first run). The witness rule is a comparator, not a proposed procedure; on naturally disputed proofs it found what the grader found (§6). After submission the author ran more arms on the same texts (App. S, outside this paper’s protocol): later grade runs, in which the strict mark’s reversal always recurs, the flag’s not always, and a no-repair instruction that keeps the 5–6 band did not lower the pass mark’s acceptances; a re-run of the witness gate that reproduced its decisions; and a witness arm without the scheme; none settles which rule is preferable.

6 Where the Design Came From: The Campaign Audit

Failures. The study grew out of an audit (all failures: App. C, Table 2) of our campaigns (App. B) and a public one [Das, 2026]. A Nemotron P1 draft that passed two of our campaign gates claims that a positive quantity starting above one and never increasing stays above one, which the sequence 2, 1 refutes. The public GPT-5.6-Sol default P6 run uses a false lemma on minimal finite transversals.

Structural closure. $S(t) \in \{0, 1, 2\}$ counts the route’s two load-bearing obligations for which a checkpoint writes an all-instance mechanism (App. F). Model raters agree on it on the trajectory corpus (Cohen’s κ 0.78–0.84 for 4 pairs of 4 raters, 84–156 texts each); on the planted texts, in one run, Claude Opus 5 and GLM-5.3 agree on every text (κ 1.00) yet each rates 6 of the 8 fatal edits $S=2$, and GPT-5.6-Sol departs from both (κ 0.42 with each): agreement is reliability, not validity; S scores written mechanisms and was never validated against an expert. It is route-specific (10 of 12 P2 lineages fall outside its route; a lineage, one solving run’s checkpoints, has no replicate).

Evaluator variation. Same-text repeats rarely change S (Claude Opus 5 on 84 repeatedly rated corpus texts: 14 of 504 same-text pairs differ, on 4 texts), and variation that could masquerade as progress rarely does when a lineage is re-rated; the alarm that flags evaluator disagreement is retrospective except on the planted texts (App. I).

Witness discovery. A prespecified comparison (App. J) ran a witness procedure and an ordinary grader on 20 natural texts, with two other models judging each objection. Both recovered every documented defect (6 distinct defects on the 7 texts that have one) and substantiated local errors on the same 4/7 texts in open dispute. The arms were matched in call count and checking rule, not in information (the witness prompts carry the marking scheme).

Reference adjudication. Every objection was then adjudicated against a ledger of required statements from Chen [2026], produced by a model under the author’s direction and reviewed by the author (App. K); here the checkers, GPT-5.6-Sol and Claude Opus 5, are the objection-producing labs. Endorsement of the 96 objections (95/96 by the same-lab checker, 95/96 by the other) checks an objection’s local claim, not whether it finds a defect. The author compared our campaign’s six GPT-5.6-Sol proofs (4 of them planted bases) with the notes and found them consistent (not a line-by-line or expert check; no written record was kept; App. P adds two model comparisons).

7 What an Operator Can Do; Limitations; Release

Operator. Know where the pass mark sits in the rubric’s bands; state the convention before comparing graders; read the objections, not only the score; under a strict rule, route two labs’ disagreements to a person, but treat agreement as no certificate (under the pass mark both accepted 13 of the 16 broken texts in both runs); the effort a referral takes was not measured. *Threshold verification* (a human certifies each load-bearing obligation, a gate the rest) is our unmeasured suggestion for races whose output goes unread [Tao, 2026].

Limitations. No independent mathematician reviewed any base, label, objection or rating; the ledger, recovery judgments, adjudication notes and pilot reads, previously credited to the author (App. Q), were produced by a model under the author’s direction and reviewed by the author (App. P). Base acceptance is model verdicts, same-lab for five bases; Claude models, one grader’s lab, wrote the edits, their intended labels and the checks, which certify falsity only, so Claude’s quoting and its lead under RT may be same-lab effects; five labels are disputed. The 32 texts are non-independent: 8 hand-picked bases, four problems (3 for the undisputed broken texts), 3 authoring models (six bases by GPT), 2 graders (a third rater, GLM-5.3, on closure only), one rubric. The witness and closure arms are reference-guided and ran once in this protocol; no head-to-head comparison with RefGrader or ProofBench’s graders and no authorship-reversed bases were run. An expert regrade of even ten texts would anchor the rest (further limitations: App. E).

Release. Third-party texts: reproduction only (NOTICE); the rest: CC BY 4.0. Code, data and model responses: <https://github.com/bobbercheng/verification-gap>.

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A Prior Campaign: IMO 2025 with an Open-Weight Model

Before IMO 2026 we reproduced the verification-and-refinement pipeline of Huang and Yang [2025] with gpt-oss-120b [OpenAI, 2025] on IMO 2025. Three observations shaped the 2026 protocol.

Effort and search. Breadth-first best-of-3 generation with model-based verification at high reasoning effort produced verifier-accepted solutions for Problems 1–5; low and medium effort did not. Tree search (MCTS) showed no advantage at the sample sizes we could afford, and an adversarial-critic refinement loop exhibited *answer lock*: it strengthened proofs of a wrong answer instead of revisiting the answer, timing out on every test case.

Verifier integrity. An externally reported low-effort success of the pipeline traced to a verifier bug that accepted invalid solutions; with the verifier fixed, no problem was solved at low effort. This incident motivated the separate second-order dependency audit of Appendix B and the negative-control discipline we recommend for any model-graded evaluation.

Tool-assisted conjecturing. On Problem 6 every search method produced plausible but wrong constructions. Enumerating small cases with a CP-SAT solver [Perron and Didier, 2023] ($n = 4: 5$ tiles, $n = 9: 12$, $n = 16: 21$) and asking the model to induce a closed form yielded $f(n, k) = n + 2k - 3$ and the correct final value 2112 for $n = 2025$, $k = 45$, in 38 seconds. This is conjecture generation from verified data, not a proof: the lower bound remained unproved, and three points constrain a formula only through the model’s prior over “nice” expressions.

B Rubric and Gate Text

The texts below are the prompts sent on 2026-07-20, copied from retained GPT-5.6-Sol invocation records with absolute run-directory paths abbreviated to <run>/. They contain no mathematics specific to any problem: the same reviewer and verifier text was used for all six problems of the GPT-5.6-Sol campaign (the other campaigns ran the same loop; Appendix C), and the only problem-specific input is the statement file. The gate is therefore portable by construction; the body’s failure catalogue shows what such a gate cannot catch without problem knowledge or a witness. The reviewer and verifier run in fresh contexts. The verifier sees the structured review, but not the reviewer’s private reasoning. These prompts request file operations; they did not restrict the available toolset (Appendix E).

Reviewer-swap adaptation. The later swaps use tool-free APIs and published proofs, without the original candidate manifests or a second-order audit. Their prompt retains the four rubric dimensions and the host pass predicate, replaces file operations with inline text and JSON output, explicitly requires empty `unverified_claims`, and tells reviewers to put checked identities in `verified_claims`. It also forbids using published solutions. This is an adapted prompt, not a verbatim replication; the bundle includes its exact text and per-request hashes.

Attack role (adversarial review).

EXECUTE THIS REVIEW NOW. Do not ask what the user wants and do not stop after reading files.

You are a fresh adversarial reviewer, not the author.

Read the task at <run>/statement.md, the proposed result at <run>/candidate_v1.md, and its manifest at <run>/candidate_v1.json. Re-derive the critical steps independently. Search for counterexamples, hidden quantifier changes, unjustified limit/continuity claims, degenerate cases, circular arguments, and gaps between experiments and proof. Do not modify Claude Code configuration files.

You MUST call the Write tool to create <run>/review_v1.json as JSON with exactly these fields:

- task_id: 'imo-2026-p1'
- verdict: one of "pass", "revise", "reject"
- scores: object with correctness, completeness, rigor, self_contained; each integer 0..4
- fatal_issues: list
- required_fixes: list
- verified_claims: list
- unverified_claims: list
- summary: concise assessment
- transfer_assessment: list of reusable or misleading mechanisms

Use "pass" only when there is no known fatal gap and every score is at least 3.

This is still a model review, not an official proof certificate.

Before ending, use Read to confirm that <run>/review_v1.json exists and contains valid JSON. Your final response must be REVIEW_ARTIFACT_COMPLETE only. Do not return the JSON merely as chat text.

Gate rule (host code, not model-visible). A review clears the gate only if the host predicate below holds; the model's verdict alone is never sufficient. In particular a "pass" that lists any unverified_claims is rejected and triggers a revision cycle, which is what happened to the first GPT-5.6-Sol candidate on Problem 2.

```
def review_passes(review):
    return bool(
        review["verdict"] == "pass"
        and not review["fatal_issues"]
        and not review["unverified_claims"]
        and all(score >= 3 for score in review["scores"].values())
    )
```

Verify role (second-order dependency audit).

EXECUTE A SECOND-ORDER AUDIT NOW. You are not the author and your job is to try to overturn a prior pass, not summarize it.

Read <run>/statement.md, <run>/candidate_v1.md, and <run>/review_v1.json.

Reconstruct the proof or technical dependency graph. Identify the single weakest inference, reverse every inequality implication, test existence/nonzero conclusions, check circular dependencies, and seek a concrete counterexample. Agreement with the first reviewer is not evidence.

You MUST call Write to create <run>/verification_v1.json as JSON with exactly:

- task_id: 'imo-2026-p1'
- verdict: "confirm" or "overturn"
- weakest_inference: nonempty string
- counterexample_search: list
- dependency_gaps: list
- required_fixes: list
- summary: string

Use "confirm" only if dependency_gaps and required_fixes are empty after the attack. Then Read the JSON back. Reply VERIFICATION_ARTIFACT_COMPLETE only after it exists. Do not ask a question or return the JSON only in chat.

The host accepts a verification only if `verdict == "confirm"` and both `dependency_gaps` and `required_fixes` are empty. A third, line-by-line implication audit with the same fail-closed shape was added on 2026-07-21 for the Nemotron campaign after its Problem-1 draft passed both gates with an invalid local inference (Appendix C); it was not part of the GPT-5.6-Sol acceptances.

Planted-study prompts (6 September 2026). The planted-defect study (Appendix L) reads three rules from one grading response and runs two further gates with other system prompts. The grading prompt is reference-free: it carries the problem statement and the coordinator scale, asks the grader not to consult or recall a published solution or use a fixed list of required lemmas (which cannot remove what a model already knows), and names the complete field without defining it, while its scale calls 7 “complete and correct” and 5–6 “essentially complete”. The witness and closure gates are reference-guided: their prompts carry the problem’s closure rubric, a marking scheme derived from the published solution that lists the steps a complete proof needs and, on some problems, gives the answer and names example wrong approaches (Appendix L.6). The reviewed version’s description of these two gates is corrected here (Appendix Q, row Q14). Printed below, verbatim from the stored requests, are the grading prompt’s first line and instruction block, the witness gate’s instruction block (which replaces the rubric’s return block), and the P1 and P5 closure rubrics; the P3, P6 and P2 rubrics are in Appendix F.

Grading prompt (grade gate; stored request.json): its first line, then the problem statement (not reproduced), then the instruction block, identical on every problem

You are an IMO coordinator grading ONE write-up for IMO 2026 Problem 1 on the standard 0-7 scale.

[the official problem statement follows here]

```
## Instructions
Grade only what is written in the text below. Do not consult, recall, or compare against any published
solution, and do not use
a fixed list of required lemmas: judge the write-up on its own terms as a proof attempt, as a coordinator
would.
Scale: 7 = complete and correct proof; 5-6 = essentially complete, with minor gaps or a small error that
is straightforwardly
repairable; 3-4 = substantial correct progress with a major part missing or unjustified; 1-2 = minor
correct progress
(e.g. correct answer or reduction with the main argument absent); 0 = no meaningful progress, or the text
is not a proof attempt.
Numerical checks, conjectures, and claims left "to do" earn no credit. A wrong final answer caps the grade
at 1.

Return ONLY a JSON object:
{"grade": 0|1|2|3|4|5|6|7, "complete": true|false, "main_gap": "<=40 words or 'none'",
 "errors": [{"description": "<=80 words: the step concerned and what is wrong or missing", "quote":
 "<verbatim excerpt of the write-up this concerns, or null>"}, ...],
 "notes": "<=60 words">}
List at most three errors, the most severe first; an empty list means you found none.
```

Witness-gate instructions (the part of the witness prompt that follows the marking scheme)

TASK: obligation audit with witnesses. Audit the write-up obligation by obligation. The three obligations are the rubric fields "reduction", "lower_bound" and "upper_bound" (use the field mapping above for this problem). For each obligation return exactly one status:

- "written_and_valid": the text contains an argument covering all cases and you found no error in it.
- "written_but_invalid": the text presents an argument intended to cover all cases, but a specific step is false or does not follow.
- "not_written": the text does not present an all-cases argument for this obligation (absent, named or sketched only, checked only on examples, proved only under an unproved extra assumption, or deferred).

For every obligation whose status is not "written_and_valid" you must give a WITNESS with three parts:

- "quote": a verbatim excerpt of the write-up, 40 to 400 characters, copied exactly (same characters, no paraphrase, no ellipsis), containing the false claim or the inference that does not follow (written_but_invalid), or the closest the text comes to the obligation / the place where it defers it (not_written). The quote is matched mechanically against the text.
- "witness": for written_but_invalid, either an explicit COUNTEREXAMPLE (concrete objects that satisfy the hypotheses of the quoted claim and violate its conclusion, with the verification spelled out) or an explicit NON-SEQUITUR (the premises the text has established, the conclusion it draws, and why the conclusion does not follow); for not_written, the precise missing statement (what would have to be proved, for all cases) and why the quoted passage does not prove it.
- "load_bearing": one or two sentences on why the obligation depends on this step as the text is written.

For "written_and_valid" set quote, witness and load_bearing to null.

Judge only what is written. Do not use the length of the text, its confidence, or claims of completeness. Do not repair the argument for the author: if a step could be fixed but is not proved in the text, the status is not_written (or written_but_invalid if what is written is false).

Return ONLY a JSON object:

```
{
  "obligations": {
    "reduction": {
      "status": "...",
      "quote": "...",
      "witness": "...",
      "load_bearing": ...
    },
    "lower_bound": {
      "status": "...",
      "quote": "...",
      "witness": "...",
      "load_bearing": ...
    },
    "upper_bound": {
      "status": "...",
      "quote": "...",
      "witness": "...",
      "load_bearing": ...
    }
  },
  "S_valid": <number of lower_bound and upper_bound obligations with status written_and_valid>,
  "wrong_answer_stated": true|false,
  "notes": "<=60 words>"
}
```

Problem 1 closure rubric (marking scheme given to the witness and closure arms)

You are an impartial grader of partial mathematical write-ups for IMO 2026 Problem 1:

"There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m,n)$ and $\text{lcm}(m,n)/\gcd(m,n)$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius."

A complete proof needs: (E) the ELEMENTARY move description: for every prime p , a move replaces the p -adic valuations (x, y) of the two chosen numbers by $(\min(x,y), |x-y|)$, i.e. $v_p(\gcd) = \min(x,y)$ and $v_p(\text{lcm}/\gcd) = |x-y|$, and a move never outputs two 1s (if $\gcd(m,n) = 1$ the second output is $mn > 1$, otherwise the first output is $\gcd > 1$); (T) TERMINATION AND UNIQUENESS, valid for EVERY sequence of moves: an argument that no infinite sequence of moves exists (e.g. a monovariant: the product of the board never increases and strictly decreases when $\gcd > 1$, while the number of 1s strictly increases when $\gcd = 1$; or a well-founded lexicographic pair), together with the fact that when no move is possible EXACTLY one integer exceeds 1 (at most one because two integers > 1 would allow a move; at least one because every move leaves at least one output > 1 , so the board never becomes all 1s); (I) INVARIANCE, valid for EVERY move and EVERY prime p : $\gcd(v_p(t_1), \dots, v_p(t_{2026}))$ over the current board is unchanged by a move (via $\gcd(x, y-x) = \gcd(x, y)$, with the cases $x = y$ and zero valuations handled), hence $v_p(M)$ equals the $\gcd$ of the initial valuations and $M = \prod_p p^{\{\gcd_i v_p(a_i)\}}$ is determined by the initial board.

Grade ONLY what is written in the text. Do not credit claims that are asserted, conjectured, checked numerically, or left as "to do". A mechanism scored 2 must be written out as an argument for all cases; a scored 1 mechanism is named/sketched, checked only on examples, or proved with a gap (e.g. termination proved but the "at least one number stays > 1 " step missing or circular; invariance proved for the generic case but the equal-valuation or zero-valuation case unaddressed); 0 means absent or a wrong approach (e.g. relying on a false lemma such as "the product of the board strictly decreases at every move" or "a quantity that starts above 1 and never increases stays above 1").

Return ONLY a JSON object:

```
{
  "answer": 0|1|2,
  "reduction": 0|1|2,
  "lower_bound": 0|1|2,
  "upper_bound": 0|1|2,
  "lower_bound_mechanism": "<short name or 'none'>",
  "upper_bound_mechanism": "<short name or 'none'>",
  "wrong_answer_stated": true|false,
  "notes": "<=60 words>"
}
```

Field mapping for this problem: "answer" = the explicit value of M , i.e. $v_p(M) = \gcd(v_p(a_1), \dots, v_p(a_{2026}))$ for every prime p (equivalently $M = \prod_p p^{\{\gcd_i v_p(a_i)\}}$), is stated (2 correct and explicit, 1 vague or only for special boards, 0 absent/wrong); "reduction" = (E); "lower_bound" = (T) termination and exactly one survivor for every play; "upper_bound" = (I) invariance of the valuation $\gcd$ s, hence M independent of the choices; "wrong_answer_stated" = true if the text states a wrong value of M or relies on a false lemma.

Problem 5 closure rubric (marking scheme given to the witness and closure arms)

You are an impartial grader of partial mathematical write-ups for IMO 2026 Problem 5:

"Let $R_{>0}$ be the set of positive real numbers. Determine all functions $f: R_{>0} \rightarrow R_{>0}$ such that $\sqrt{x^2 + f(y)^2/2} \geq (f(x) + y)/2 \geq \sqrt{x f(y)}$ for every x, y in $R_{>0}$."

The correct answer is $f(x) = x + c$ for a constant $c \geq 0$. Published solutions differ in the second half of the necessity argument (an iterative argument on orbits with a fixed-point separation lemma, or a local calculus estimate), so the obligations below are stated independently of the route. A complete proof needs: (S) the STRUCTURAL identity: for every $t > 0$, $f(f(t)) = 2f(t) - t$ (from the two inequalities at $(x, y) = (f(t), t)$, where both outer terms equal $f(t)$), equivalently every forward orbit $t, f(t), f(f(t)), \dots$ is an arithmetic progression with common difference $d(t) = f(t) - t$, and $d(t) \geq 0$ for every t ; (V) SUFFICIENCY: verification that every $f(x) = x + c$ with $c \geq 0$ satisfies BOTH inequalities for ALL $x, y > 0$ (e.g. QM-AM-GM applied to x and $y + c$, since $f(x) + y = x + f(y)$); (N) NECESSITY: a complete argument that $d(x) = f(x) - x$ is the SAME constant for ALL $x > 0$, e.g. (i) any two positive differences A and B are equal (comparing the sequences $\sqrt{x + mA}$ and $\sqrt{y + nB}$ over iterates via the right-hand inequality, or an equivalent argument) AND (ii) the mixed case is excluded: $f(x) = x + D$ and $f(y) = y$ force $|x - y| > D$, so no function with both fixed and non-fixed points satisfies the system; or a rigorous local estimate $f(y) = f(x) + (y - x) + O((y - x)^2)$ valid for every x , giving $f' = 1$ everywhere. The case of fixed points (including f being the identity) must be handled, not assumed away.

Grade ONLY what is written in the text. Do not credit claims that are asserted, conjectured, checked numerically, or left as "to do". A mechanism scored 2 must be written out as a complete argument for all x, y ; a scored 1 mechanism is named/sketched, proved under an unproved extra assumption (continuity, monotonicity, differentiability, surjectivity, injectivity, or "f has no fixed points"), or covers only part of the cases (e.g. only the non-fixed-point case, or only one of the two inequalities in the verification); 0 means absent or a wrong approach (e.g. concluding that $f(x) = x$ is the only solution, or deducing $f(x) = x + c$ from $f(f(x)) = 2f(x) - x$ alone).

Return ONLY a JSON object:
 {"answer": 0|1|2, "reduction": 0|1|2, "lower_bound": 0|1|2, "upper_bound": 0|1|2,
 "lower_bound_mechanism": "<short name or 'none'>", "upper_bound_mechanism": "<short name or 'none'>",
 "wrong_answer_stated": true|false, "notes": "<=60 words">}
 Field mapping for this problem: "answer" = the solution set $f(x) = x + c$, $c \geq 0$, stated as the complete answer (2 explicit and correct including the range of c , 1 partial, e.g. only $f(x) = x$, or the family stated as a conjecture, 0 absent/wrong); "reduction" = (S); "lower_bound" = (V) sufficiency for all x , y and all $c \geq 0$; "upper_bound" = (N) necessity: constant difference for all x including the fixed-point case; "wrong_answer_stated" = true if the text states a wrong solution set (e.g. only $f(x) = x$) or relies on a false lemma.

C Per-Problem Notes and Artifact Manifest

Table 2: Verification failures found in the audit; each row is a released artifact. Three further cases (a gate refusal for unverified manifest claims; a wrong cross-lab dissent on GPT’s P2; two identical texts with different labels) are in Appendices C and F.

Event	Accepted by	Caught by	Consequence
Nemotron P1: nonincreasing product “stays above one”	attack review 4/4/4/4 and verify (confirm)	third line-by-line logic audit, next day	published proof edited in one sentence; fresh blind attempt passed all three
GLM-5.2 P2: decisive identity $\text{num}(E) = Q \cdot \text{num}(G_2)$ in external CAS files	gate with workspace access (4/4/3/4, confirm)	Kimi K3 tool-free re-grade (fatal: certificate not in text)	gate pass $\neq$ self-contained proof
Public GPT default P6 (false finite-transversal lemma, two checkpoints) and P3 (both mechanisms sketched)	structural annotators: $S=2$ “complete rigorous proof”	public verifier 1/7 and 2/7 (counterexample; “unverifiable sketch”)	written $\neq$ valid; validity layer $S=1$ and $S=0$
Public Kimi P6 round 2: self-dual-clutter theorem applied to a possibly infinite transversal	structural annotators $S=2$	public verifier 3/7 (fatal); fixed after critique	same

Editorial repair. The accepted Nemotron Problem-1 proof replaces one local sentence: the draft inferred that a nonincreasing product stays above one; the sufficient argument is that every legal move retains a nonunit. Editing fixes typography and documented local gaps and never converts a rejected candidate into an accepted result.

Table 3 lists, for every accepted proof, the lineage that produced it, the number of attack/revise cycles, the rubric scores of the accepting review, the verifier verdict, and the first 16 hex digits of the SHA-256 digests of the candidate, review, and verification artifacts. Full digests for every file in the supplementary bundle are in its MANIFEST. json. The six published GPT-5.6-Sol proofs are byte-identical to the accepted run candidates.

Table 3: Artifact manifest for accepted proofs. Scores are correctness/completeness/rigor/self-containedness of the accepting review. Digests are SHA-256 prefixes of the run artifacts.

Route	P	Lineage	Cycles	Scores / verify	Candidate	Review	Verification
GPT-5.6-Sol	1	efd2737d	1	4/4/4/4, confirm	cdc88080cd219cde	be0c06f7334e3089	85689a90d97f5c08
GPT-5.6-Sol	2	3c91a536	2	4/4/4/4, confirm	eb189909d1bd39a5	ed46e718c6dbca35	43e95c9fa5f8bcfc
GPT-5.6-Sol	3	ffa748a9	1	4/4/4/4, confirm	cb1bcb2df8fa81b	50ee118353e5c1eb	d7a817dced330831
GPT-5.6-Sol	4	247f5b7c	1	4/4/4/4, confirm	4d0a20e17bde3191	0c3ed2d97b8663ac	bb932d8de618b006
GPT-5.6-Sol	5	e26aacad	1	4/4/4/4, confirm	3090f2134de489ae	e160a9286b0200cc	33bc43ad338e5f1a
GPT-5.6-Sol	6	3acd733f	1	4/4/4/4, confirm	d31e74d07a68c601	50750125557778fb	045730c351052b59
Nemotron 3U	1	2853f8e6	1	4/4/4/4, confirm	1a10db6e23dca33d	eafd1f97c6a2cd84	631e3fa409d58dc1
Nemotron 3U	1*	dd064edd	1	4/4/4/4, confirm	f34c81fda467c0c6	b03adf252b32f492	9dc3e2d46672c524
Nemotron 3U	5	f3bd351b	1	4/4/4/4, confirm	28cf6b5ebb024dc2	cbf2850544e3b88a	a200369d810a6368

Per-problem notes. *P1.* Single cycle; the reviewer confirmed the (P, K) descent and the Euclidean invariant on p -adic exponents. *P2.* The first candidate (4ed2348c...) passed the reviewer 4/4/4/4 but listed two unverified manifest claims—an undocumented symbolic factorization and random numerical tests—so the host gate refused it; cycle 2 removed the nonessential claims and the cleaned candidate cleared review and verification. This was evidence hygiene, not a proof repair. *P3.* The first lineage (5206d120) timed out after writing a candidate and was not reviewed; the second lineage inherited it and passed both gates in one cycle. *P4–P6.* Single cycle each. GLM-5.3’s reviewer-swap dissent on P2 objects to a directed-angle step that is in fact immediate: E lies on line CA , so $\angle FCE = \angle(CL, CA)$.

Nemotron P1. The 2026-07-20 acceptance (2853f8e6) passed review and verification, but the line-by-line logic audit added on 2026-07-21 overturned it on one local inference (a nonincreasing product was said to stay above one). A fresh blind attempt the same day (dd064edd*) produced a corrected proof that passed review, verification, and the logic audit (3f166924. . .). The published Nemotron P1 text is the first draft with that single sentence replaced, as disclosed above. *Nemotron P5*. Accepted in one cycle after two earlier lineages were rejected; its later logic audit also confirmed (f587f25e. . .). Published Nemotron copies are typographically edited, so their digests differ from the run artifacts listed here; the bundle carries both.

Table 4: Nemotron 3 Ultra rejections. Scores are the best review scores over all attempts (correctness / completeness / rigor / self-containedness, each out of 4).

P	Best scores	Decisive defect
2	1/1/1/2	Spiral-similarity centers at L and K inferred from angle equalities at other vertices; the conditions at the centers were never derived, so later coordinate claims had no base.
3	1/1/1/1	Correct value $2^n / (2^{n+1} - 1)$ stated; upper-bound cuts did not span the stick for $n \geq 2$, the lower bound failed at $n = 2$, both refinement lemmas unproved.
4	1/1/1/2	Strategies for 60° , 90° , and above sound; the claimed defensive cycle below 60° fails at 30° ; 60° – 90° lacked the exhaustive invariant.
6	0/0/1/2	False premise that each next term’s prime support is a minimal hitting set (6, 8, 10 refutes it); the CRT construction yielded a coprime, not intersecting, number. Two revisions did not repair the foundation.

D Quote Localization

A decision is easier to check when its evidence names a place in the proof. `scripts/inspection_cost.py` measures that from the stored objections with no model calls. A *unit* is a heading-delimited section, the median proof having 8; every anchored quote is mapped to the unit containing it. On the 16 fatal-or-gap texts the ordinary grader’s error quotes give a pointer on 16/16, covering 2 sections of the median proof, and the witness objections give one on 16/16, covering 1; over all 24 edits the counts are 24/24 and 21/24 with medians of 1 and 1 (Table 5). Both rules therefore hand a reader a place to look, and they differ in how tightly they point.

This is quote localization and nothing more. Locating a quote does not establish that the sections it does not touch are correct, so these counts are not estimates of verification work saved, and no claim in this paper rests on them. An earlier version of this analysis also reported dependency-closure sizes, downstream consumers, a prose-localization rate for the completeness verdict and a share of objections carrying a finite check; those rested on heuristics that mis-parse decimal equation labels, treat a lemma’s mention as its definition, absorb prose into a generic heading and match the ordinary word “set”, and were withdrawn rather than repaired (DEVIATIONS.md).

Table 5: Where each rule’s evidence points, by population: how many texts receive at least one anchored quote, and the median number of distinct sections those quotes fall in among the texts that receive one.

Evidence the rule returns	fatal or gap ($n=16$)		all edited ($n=24$)	
	with a pointer	sections	with a pointer	sections
ordinary grader’s error quotes	16/16	2	24/24	1
witness objections	16/16	1	21/24	1
witness objections, reference-adjudicated	16/16	1	21/24	1

E Network-Access Audit

Limitations. The paper’s limitations are stated in the body (§7); in the reviewed version they sat here, under this appendix’s title. The appendix keeps its title and subject, the network-access audit below, and the further limitations that the body’s list does not repeat.

Further limitations. The disagreement alarm is retrospective everywhere except on the planted texts. The completeness verdict is the more lab-dependent of the two fields of a grading response: in the first run the labs’ completeness verdicts differ on 10 of the 24 mutants and on 0 of the 8 originals, and their pass-mark decisions on 2 of the 24 mutants. GPT-5.6-Sol also wrote five of the twenty witness texts. The closure rubric scores a wrong approach 0, so *S* measures written-mechanism adequacy rather than syntax alone. Checkers call almost every objection’s local claim correct whichever lab produced it (same-lab 95/96, other-lab 95/96; the one exception was rejected by both); endorsement is not a verdict that the objection identifies a defect, so it shows no same-lab effect and does not rank objections. Seven adjudication statuses split with no human resolution. Lineages are single unmatched runs. Public grades are model grades (the public campaign’s Claude-based verifier agents), not coordinator marks. The ledger as used has two domain omissions, corrected in `LEDGER_v2.md`; no judgment used the corrected edition and no scored outcome depends on either correction, but the effect of showing models the corrected ledger is unmeasured, and a model check of 29 September 2026 found two further ledger issues that change no recorded judgment (Appendix K). The planted-defect comparison reads acceptance rules off one set of grading responses; it does not measure whether a model can find natural defects, and its intended labels were written by AI agents to the author’s design (Appendix L.1).

Solver prompts forbade retrieving published solutions, but enforcement differed by route. The July 20 Codex route bypassed sandboxing and approvals; omitting a web-search flag did not remove its web tool. Nemotron’s initial acceptances used Claude Code’s default toolset, including WebSearch and WebFetch; those tools were removed on July 21. GLM-5.2 used restricted file/shell tools and, in later blind invocations, a hook rejecting network-capable command patterns. That hook was a software guard, not OS-level network isolation. Kimi K3’s shell used a container with networking disabled; GLM-5.3’s direct messages requests exposed no tools to the model.

The retained-stream audit covered 25 Codex, 216 Nemotron, and 1,865 GLM-5.2 streams. It found no Claude WebSearch/WebFetch calls and no network-capable shell commands, but one Codex `web_search` page-open event: the P5 reviewer opened the official statement URL supplied in its task during 21:29–21:32 UTC on July 20. No additional query or solution retrieval was observed. The returned page body and two nested Nemotron agents’ inner calls were not retained, so this is an observation about logged behavior, not proof of isolation or absence of contamination.

F Structural Closure: Corpus, Annotation Layers, Adjudication, Agreement

Sample and units. The corpus has 176 records (156 distinct input hashes) in 41 lineages, reproducible from the released annotations by `scripts/closure_matrix.py`. Problem 3: 57 records in 17 lineages—20 local discovery snapshots, two local repair outputs, nine public final files, and 26 `write_file` `current.md` snapshots reconstructed from the public logs. Problem 6: 42 records in 12 lineages (14 local, nine public finals, 19 public snapshots). Problem 2: 77 records in 12 lineages (11 local, nine public finals, 57 public snapshots). A public lineage is one model at one effort setting; our lineages aggregate attempts and harnesses and contribute only retained candidate files, so attempts that left no candidate count toward budgets but have no checkpoint (GLM-5.2’s 24 reviewed P3 attempts left four retained candidates). The two Problem-3 repairs descend from Kimi R1. Records are the retained texts, not an exhaustive census of edits or reasoning states: shell rewrites and files other than the tracking file are not observed, so a crossing located “inside one context” is bounded by sparse writes.

Five label layers. Every distinct text carries five labels, all released in `closure_adjudicated/labels.json` and produced by `scripts/closure_adjudicate.py`. *raw*: layer 0, GLM-5.3 (Z.ai, tool-free) on each of the 176 records with the per-problem rubric below and the checkpoint text alone, without author label or workspace; the script enabled thinking and retried without thinking if no JSON was parsed, which happened for 61 of 176 records (2/57, 18/42, 41/77 on Problems 3, 6, 2), always because the thinking call stopped at its output cap; where a text recurs under several record ids, the raw label is the majority over its records. *glm*: a GLM-5.3 re-rating of each of the 156 distinct texts with thinking and a larger cap (9 of 156 still fell back to a no-thinking call; all 156 re-ratings parsed, so the layer-0 fallback the script allows for a missing re-rating was never used). *kimi*: Kimi K3 (Moonshot, tool-free, same rubric) on each distinct text. *written*: the conservative aggregation of *glm* and *kimi*, field by field: a mechanism counts 2 only if both annotators score it 2,

Table 6: Valid-layer S of the last checkpoint/grade. Public: nine runs of one thin harness [Das, 2026], grade = verifier first-pass grade (“ $a \rightarrow b$ ”: repaired later); ours: gate outcome. xhigh/Pro/max/default/ultra: GPT-5.6-Sol effort settings; R1/R2: our Kimi runs; * $S=2$ with grade < 5 ; † wrong answer; ‡ non-synthetic route (rubric N/A).

Model	P2	P3	P6
Claude Fable 5	2 / 7	2 / 7	2 / 7
GPT-5.6-Sol xhigh	2 [†] / 4 $\rightarrow$ 7	2 / 7	2 / 7
Kimi K3	1 [†] / 7	2 / 5 $\rightarrow$ 7	2 / 3 $\rightarrow$ 7
GPT-5.6-Sol Pro	1 [†] / 7	0 / 2	2 / 7
GPT-5.6-Sol max	0 [†] / 7	0 / 1	0 / 1
GPT-5.6-Sol default	0 [†] / 4	0 / 2	1 / 1
Muse Spark 1.1	0 [†] / 2	0 [‡] / 1	1 / 2
DeepSeek V4 Pro	0 [†] / 2	0 [‡] / 1	0 / 1
Grok 4.5	0 [†] / 2	0 / 2	0 / 1
GPT-5.6-Sol ultra	2 / pass	2 / pass	2 / pass
Kimi K3 R1	–	2 / 6 $\rightarrow$ 7	–
Kimi K3 R2	–	0 / 1	–
GLM-5.2	0 [†] / pass	0 / 2	2 / pass
GLM-5.3	–	0 / unreviewed	–
Nemotron	0 [†] / fail	0 [‡] / fail	0 [‡] / fail

otherwise it takes the lower score; if either rating were missing the label would be *unknown*, never a silent fallback to layer 0 (at freeze all 156 texts have both ratings and none is unknown). *valid*: the written label with independently refuted mechanisms set to 0 (next paragraph). S counts the lower_bound and upper_bound fields scored 2; the preparatory field (reduction) and answer are recorded but not counted. The written layer is a model-score aggregation convention, not an expert adjudication, and the paper calls it that. Trajectory counts use the defect-corrected layer with the written layer alongside; Table 7 gives the headline counts under all five layers.

Validity criterion. A mechanism scored 2 by the instrument is set to 0 in the defect-corrected layer when an independent record shows that the written argument for it is invalid as written: an explicit counterexample to a lemma it relies on, or a fatal invalid step in that mechanism recorded by the public verifier or by our gate. The criterion is applied to every retained text carrying the same defect, not only to final files, and no entry rests on the author’s mathematical judgement alone: closure_adjudicated/validity_overrides.json stores the criterion and, per entry, the reason and the record it cites. There are no author overrides. Three entries are in force, all on the Problem-6 finiteness mechanism. Two public GPT-5.6-Sol default texts (main-session turn 9 and the final file) rest on the lemma that every pairwise-intersecting family of finite sets has finitely many inclusion-minimal finite transversals, proved once by induction and once by a König-tree argument. The lemma is false: with distinct symbols, $F_n = \{x_1, \dots, x_n, y_n\}$ ($n \geq 1$) is pairwise intersecting (every member contains x_1), yet $T_k = \{y_1, \dots, y_{k-1}, x_k\}$ is an inclusion-minimal finite transversal for every k (it meets F_n in y_n for $n < k$ and in x_k for $n \geq k$; dropping y_j misses F_j , dropping x_k misses F_k), so there are infinitely many. The public verifier’s 1/7 grade records a machine-verified counterexample of the same kind. Both annotators credited the lemma in both texts (layer 0, the re-rating, and Kimi all score finiteness 2): written $S=2$, valid $S=1$ for both, and the $2 \rightarrow 1$ “regression” that v3 reported for this lineage, produced by correcting the final file alone, disappears. The third entry is public Kimi K3’s round-2 turn-7 file, whose finiteness proof extracts a minimal transversal by Zorn’s lemma and applies a self-duality claim proved only for finite minimal transversals, so the step is circular in the case it needs. The statement it aims at is not refuted (the lineage later proved it by a compactness argument, re-graded 7/7), but the argument as written is: the public verifier’s first pass recorded the step as the fatal defect (3/7), and the harness quoted it to the next session (“self-duality was only established for FINITE minimal transversals”). Written $S=2$, valid $S=1$; under the defect-corrected layer the lineage runs $0 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 2$ and the corpus has no regressing lineage. The step-by-step comparisons behind these entries, in closure_adjudicated/adjudication_notes.md, were written by a model (a Claude subagent in Claude Code) under the author’s direction and reviewed by the author.

Problem-2 route policy. The Problem-2 rubric scores the synthetic route of the published solution (a key circle through A proved to contain K and L ; $OM = ON$ from it). Before any count was regenerated, every Problem-2 lineage was classified by the route of its final retained text and both annotators’ mechanism notes (closure_adjudicated/p2_routes.json, with evidence quotes), and the classification applies to every checkpoint of the lineage. The rubric applies to two lineages:

public Claude Fable 5 (directed-angle translation of the hypotheses into concyclicities; an affine power identity vanishing at A, K, L in place of the explicit circle; finish by the median-length formula) and our GPT-5.6-Sol ultra (the circle with diameter AZ shown to contain K and L by radical axes; O the midpoint of AZ). The other ten are route N/A, displayed in Table 6 but excluded from agreement and transition counts: GPT-5.6-Sol Pro (trigonometric), GPT-5.6-Sol max (Cartesian coordinates with a polynomial certificate), GPT-5.6-Sol xhigh and default (complex coordinates), Kimi K3 (law-of-sines and power-of-a-point coordinates), DeepSeek V4 Pro (complex plan, unexecuted), Muse Spark 1.1 (coordinate reduction only), Grok 4.5 (no route executed), GLM-5.2 (sine-rule parametrisation with an external CAS certificate), and Nemotron (complex reality conditions and spiral similarity, unexecuted). Of the published material available to us only Chen [2026] covers Problem 2, with one synthetic solution; the other route classes are read off the texts, not matched to a published solution. Under this policy Problem 2 is nearly uninformative for the trajectory analysis (two lineages, both ending at $S=2$ with the crossing inside the record, no $S=1$ checkpoint), a limitation of a route-specific obligation set on a problem with several workable routes; v3’s two Problem-2 “disagreements” (GPT-5.6-Sol max at $S=0$ with grade 7, Kimi K3 at $S=1$ with grade 7) were non-synthetic proofs scored against synthetic obligations.

Reliability. Cohen’s κ on S : GLM–Kimi 0.81 ($n = 156$), GLM–Claude 0.78, Kimi–Claude 0.78 ($n = 156$), GLM–GPT 0.84 ($n = 84$, P3/P6); between the GLM-5.3 re-rating and Kimi K3 by problem: P3: $\kappa = 0.88$, exact agreement 96% on S (43/48 and 40/48 on the two obligation fields); P6: $\kappa = 0.95$, exact agreement 97% on S (31/36 and 32/36 on the two obligation fields); P2: $\kappa = 0.64$, exact agreement 88% on S (58/72 and 62/72 on the two obligation fields). Overall $\kappa = 0.81$, exact agreement 92%; of 156 texts, 141 have both annotators agreeing on S , 12 were resolved by the conservative rule, 3 carry a validity entry, 0 are unknown (a rater missing). Problem 2 is where the rubric fits few texts. Layer-0 labels of repeated texts (the same bytes retained under two record ids) agree on 17 of 19 (8/8, 6/6, 3/5 on Problems 3, 6, 2); the two conflicts are the Problem-2 GPT-5.6-Sol max and default final files against their last snapshots, which is why labels are per text rather than per record and why `closure_matrix.py` refuses to report a transition between identical texts and fails if any hash carries two labels. 11 of 156 valid-layer labels differ from layer 0 (P3 deedy-gpt-5.6-sol-final: $2 \rightarrow 0$ (conservative); P6 p6-gpt-5.6-sol-main-t009: $2 \rightarrow 1$ (validity override); P6 p6-kimi-k3-round2-t007: $2 \rightarrow 1$ (validity override); P6 p6-deedy-gpt-5.6-sol-final: $2 \rightarrow 1$ (validity override); P2 p2-gpt-5.6-sol-pro-main-t022: $1 \rightarrow 0$ (agree); P2 p2-gpt-5.6-sol-max-main-t071: $1 \rightarrow 0$ (conservative); P2 p2-deedy-gpt-5.6-sol-max-final: $2/0 \rightarrow 0$ (conservative); P2 p2-deedy-gpt-5.6-sol-final: $0/1 \rightarrow 0$ (agree); P2 p2-gpt-5.6-sol-pro-main-t038: $2 \rightarrow 1$ (conservative); P2 p2-gpt-5.6-sol-xhigh-round2-t005: $0 \rightarrow 1$ (agree); P2 p2-deedy-kimi-k3-final: $2/2 \rightarrow 1$ (conservative)). These κ values measure agreement between model raters on the trajectory corpus; on the 32 planted texts, in the planted study’s closure gate (run once), GPT-5.6-Sol and Claude Opus 5 agree on S with $\kappa = 0.42$, and S has not been validated against an expert.

Sensitivity by layer. Table 7 reports, under each layer, the counts the paper’s observations rest on: checkpoints at $S=1$, crossings to $S=2$ that lie inside the record, and lineages that lose a written mechanism. No Problem-3 checkpoint sits at $S=1$ under the GLM, written, or defect-corrected layers; under single evaluators the counts are GLM-5.3 0, Kimi K3 1, Claude Opus 5 1, GPT-5.6-Sol 2 (the round-3 turn-7 Kimi file, whose lower bound Kimi, Claude, and GPT score 2 and GLM 1; next paragraph), and the two-of-three majority gives 1. The defect-corrected layer adds the three refuted Problem-6 texts to the $S=1$ count and removes the only regression; crossings inside the record stay at four or five on Problems 3 and 6 in every layer (two, on the two synthetic Problem-2 lineages).

Reference-anchored adjudication. Where a single label decides an observation, the text was compared step by step with a published argument or a recorded verifier rationale (by a model, a Claude subagent in Claude Code, under the author’s direction, and reviewed by the author; the author does not act as an expert, and no independent mathematician checked the comparisons, which are released in `closure_adjudicated/adjudication_notes.md`). (i) The one Problem-3 text on which the annotators split, public Kimi’s round-3 turn-7 tracking file, contains every step of the published tree-component lower bound [Chen, 2026] in outline: a multigraph on the pieces, a count forcing a tree component, the bipartite-sign inequality, and a proved estimate for super-increasing pieces. It asserts one supporting claim the published proof does not need (a min-cost matching formula, “by an exchange argument”) and labels itself a sketch “modulo write-up”. Under the rubric’s written-out standard it is $S=0$ (written and defect-corrected layers); under Kimi’s reading $S=1$. Read

Table 7: Headline counts under each label layer. P2^s: the two synthetic-route Problem-2 lineages only.

Layer	rater(s)	$S=1$ checkpoints			crossings in record			regressing lineages		
		P3	P6	P2 ^s	P3	P6	P2 ^s	P3	P6	P2 ^s
raw	layer 0 (GLM-5.3, every checkpoint)	0	4	0	5	5	2	0	1	0
glm	GLM-5.3 re-rating	0	4	0	5	5	2	0	1	0
kimi	Kimi K3	1	5	0	4	4	2	1	1	0
claude	Claude Opus 5	1	6	0	5	4	2	1	1	0
gpt	GPT-5.6-Sol (P3/P6; authored some texts)	2	7	–	4	4	–	1	0	–
written	GLM $\cap$ Kimi (conservative)	0	4	0	4	5	2	0	1	0
majority3	2 of 3 labs (GLM, Kimi, Claude)	1	4	0	5	5	2	1	1	0
valid	written, refuted mechanisms removed	0	7	0	4	4	2	0	0	0
rep1	GLM-5.3 repeat 1 (same settings)	0	–	–	5	–	–	0	–	–
rep2	GLM-5.3 repeat 2	–	–	–	–	–	–	–	–	–
rep3	GLM-5.3 repeat 3	–	–	–	–	–	–	–	–	–
crep1	Claude Opus 5 repeat 1	1	5	–	4	5	–	0	1	–
crep2	Claude Opus 5 repeat 2	1	4	–	4	5	–	0	1	–
crep3	Claude Opus 5 repeat 3	0	6	–	5	4	–	0	1	–

against the reference, the lower bound was in hand in outline one session before both mechanisms were written out (round 4), so the absence of $S=1$ states on Problem 3 is partly an artifact of the write-up standard; the record cannot separate “not found” from “found but not written to standard”. (ii) The Problem-6 counterexample and the three refuted texts, above. (iii) The Problem-2 routes, above.

Timing and scope. 15 of 31 applicable lineages end at $S = 2$ (7/17, 6/12, 2/2 on Problems 3, 6, 2); 10 crossings lie inside the record (4, 4, 2); 0, 7, and 0 checkpoints sit at $S = 1$; 0 lineages regress (defect-corrected layer; the other layers in Table 7; layer-0 values are recoverable from the released annotations). Public GPT-5.6-Sol default on Problem 3 is $S=0$ at turn 11 and its final file is $S=0$ in the written and defect-corrected layers: layer 0 and the GLM re-rating score both mechanisms 2, Kimi scores both 1 (sketches), and the public verifier graded the file 2/7 as an “unverifiable sketch”; the lineage never crosses. Public Kimi’s round-4 turn-14 snapshot is $S=2$, but later shell edits changed the frozen 5/7 candidate, and its final file follows repair. Public Kimi on Problem 6 runs $0 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 2$ in the written layer and $0 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 2$ in the defect-corrected layer: the round-2 text’s finiteness step is the refuted one, the round-3 file is $S=1$, and the next session’s first read of the tracking file is byte-identical to that $S=1$ record; its prompt contained an explicit critique of that file (Appendix N). Full-run grades may consider supporting files omitted from a single-text annotation, so Table 6 pairs run outcomes with sampled progress, not grades of identical bytes.

Critique-associated gains and repairs. Both recorded below-closure gains followed critique: GLM-5.2 on Problem 6 ($0 \rightarrow 1 \rightarrow 2$ over three reviewed revisions) and public Kimi K3 on Problem 6, which crossed to a defect-corrected $S=2$ in the session whose prompt carried an explicit critique of an $S=1$ file byte-identical to its first read (Appendix N); neither had a no-critique control. All three recorded repairs from an $S=2$ text with reviewer-localized defects reached a blind 7/7 (two seeds, one problem; one repair by GLM-5.2, which never reached $S=2$ on Problem 3 alone).

Agreement check and provenance classes. Under the route policy 31 lineages are evaluable (17, 12, 2 on Problems 3, 6, 2) and ten Problem-2 lineages are route N/A; GLM-5.3’s Problem-3 lane was never reviewed and is skipped. Public lineages use the campaign’s verifier grades; where the last record is the final file and a repair session intervened, the final grade is the stage-matched grade, otherwise the first-pass grade. Our lineages use the gate outcome of the retained review *and* verification receipt of a checkpoint where both exist, and otherwise the campaign’s recorded outcome (Kimi R1 6/7, Kimi R2 1/7, GLM-5.2 2/7, both repairs 7/7); Kimi R1’s 6/7 is a retrospective record, its frozen defect review saying *revise* with a null score, and GLM-5.2’s final simple-campaign receipt is also null-scored. The comparison rule is $S=2$ iff grade ≥ 5 or gate pass. Under the defect-corrected layer 30 of 30 evaluable lineages agree (10 on other routes N/A, 1 without a grade or label) (first-pass grades instead: 29 of 30); under the written layer the GPT-5.6-Sol default Problem-6 final ($S=2$ written, grade 1) is the one disagreement. Both tallies are in the released `regularities.json`. This is retrospective agreement between two model-derived labels of heterogeneous provenance, not a validation of transitions and not calibration of a predictor. Budget, descriptively: median completion tokens of public lineages ending at $S=2$ vs. below are P3 331k vs. 141k, P6 111k vs. 21k (valid layer; layer-0 labels gave 187k vs. 224k and 100k vs. 106k); the direction depends on the label layer

and totals mix repair and outage sessions with unmatched budgets, so no direction is drawn; large budgets did not guarantee written mechanisms (326k tokens at $S=0$).

GLM-5.3 discovery budget. The four original discovery calls used 64,000, 131,072, 131,072, and 131,072 output tokens: 457,216 total, all without candidate text. Four separate serializers extracted checkpoints from retained thinking and used 13,250 more output tokens (470,466 including serialization). The first result was recovered from its saved stream, not generated by an additional call. The bundle includes the per-call configuration and usage audit.

Problem-3 rubric (verbatim).

You are an impartial grader of partial mathematical write-ups for IMO 2026
Problem 3:

"Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1. Liu marks at most n points, then Xiang marks at most n points (all distinct); the stick is cut at all marks; they alternately claim pieces, Liu first, each maximizing own total. For each n determine the largest c such that Liu can guarantee at least c ."

The correct answer is $c_n = 2^n / (2^{n+1} - 1)$. A complete proof needs: (R) the reduction of the claiming phase to an alternating sum of sorted piece lengths; (L) an all- n LOWER bound: a Liu division $(1:2:4:\dots:2^n)$ and an argument valid against EVERY Xiang reply (e.g. a pairing/tree argument on rank pairs, an even-position majorization, or an induction that handles arbitrary interleaving); (U) an all- n UPPER bound: for EVERY Liu division a Xiang reply with at most n cuts (e.g. subset-sum pigeonhole giving a gap $\leq 1/(2^{n+1} - 1)$ plus a cancellation/overlay construction).

Grade ONLY what is written in the text. Do not credit claims that are asserted, conjectured, numerically tested, or left as "to do". A mechanism scored 2 must be written out as an argument for all n ; a scored 1 mechanism is named/sketched or proved only for specific small n ; 0 means absent or a wrong approach.

Return ONLY a JSON object:
{ "answer": 0|1|2, "reduction": 0|1|2, "lower_bound": 0|1|2, "upper_bound": 0|1|2, "lower_bound_mechanism": "<short name or 'none'>", "upper_bound_mechanism": "<short name or 'none'>", "wrong_answer_stated": true|false, "notes": "<=60 words" }
Scoring key for answer: 2 = correct closed form stated as the answer; 1 = correct values for small n only or correct form stated as conjecture; 0 = absent or wrong.

Problem-6 rubric (verbatim).

You are an impartial grader of partial mathematical write-ups for IMO 2026
Problem 6:

"Let $a_1, a_2, \dots$ be an infinite sequence of integers greater than 1 such that for all n , a_{n+1} is the smallest integer greater than a_n with $\gcd(a_{n+1}, a_i) > 1$ for every $i \leq n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n ."

A complete proof needs: (C) the characterization that an integer $m \geq a_1$ is a term iff its prime support meets the prime support of every term (equivalently

of every earlier term); (F) the FINITENESS / large-prime-erasure lemma: membership depends only on the primes in some finite set (e.g. primes $\leq a_1$ or $\leq a_1^2$), because any term divisible by a large prime is dominated by a smaller number with the same small-prime support that is itself a term -- proved for ALL terms, not checked on examples; (P) the PERIODICITY assembly: with L = product of the finitely many relevant primes, membership is invariant under $m \rightarrow m+L$ for every $m \geq a_1$ (from the first term, not merely eventually), and T is the number of terms in a window of length L , giving $a_{n+T} = a_n + L$ for all $n \geq 1$.

Grade ONLY what is written in the text. Do not credit claims that are asserted, conjectured, numerically tested, or left as "to do". A mechanism scored 2 must be written out as an argument for all cases; a scored 1 mechanism is named/sketched or checked only on examples; 0 means absent or a wrong approach (e.g. relying on a false lemma such as "each new term's prime support is a minimal hitting set").

Return ONLY a JSON object:

```
{
  "answer": 0|1|2,
  "reduction": 0|1|2,
  "lower_bound": 0|1|2,
  "upper_bound": 0|1|2,
  "lower_bound_mechanism": "<short name or 'none'>",
  "upper_bound_mechanism": "<short name or 'none'>",
  "wrong_answer_stated": true|false,
  "notes": "<=60 words"
}
```

Field mapping for this problem: "answer" = the claimed structure (L a product of finitely many small primes, T a count of residues) is stated (2 correct and explicit, 1 vague, 0 absent/wrong); "reduction" = (C); "lower_bound" = (F) finiteness / large-prime erasure; "upper_bound" = (P) global periodicity from $n=1$; "wrong_answer_stated" = true if the text proves only eventual periodicity or relies on a false lemma.

Problem-2 rubric (verbatim).

You are an impartial grader of partial mathematical write-ups for IMO 2026
Problem 2:

"Let ABC be a triangle, M and N the midpoints of AB and AC . Points K and L lie strictly inside triangles BMC and BNC respectively, with K strictly inside triangle ABL and L strictly inside triangle AKC , such that $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of AKL . Prove $OM = ON$."

A complete proof needs: (T) TRANSLATION of the three angle conditions into concyclicities (e.g. with $X = BK$ meets AC and $Y = CL$ meets AB : B, Y, X, C concyclic; X, N, L, B concyclic; Y, M, K, C concyclic) or an equivalent coordinate/complex setup that actually encodes all three hypotheses; (K) the KEY CIRCLE: an explicit identification of a circle through A that is proved to contain both K and L (e.g. the circle through A and the midpoints of BY and CX , or the circle with diameter AZ where Z is the centre of $(BCXY)$), with the membership of K and L PROVED (power of a point / radical axis / directed angles), not asserted; (F) the FINISH: $OM = ON$ derived from that circle (equal powers of M and N , or O identified as a midpoint so that MO and NO are half of equal radii), including handling of configuration/sign issues.

Grade ONLY what is written. Do not credit claims that are asserted, conjectured, checked numerically, or left "to verify". A mechanism scored 2 must be written out as a complete argument; 1 means named/sketched or reduced to an unexecuted computation; 0 means absent or a wrong approach (e.g. spiral-similarity centres inferred from angles at the wrong vertices).

Return ONLY a JSON object:

```
{
  "answer": 0|1|2,
  "reduction": 0|1|2,
  "lower_bound": 0|1|2,
  "upper_bound": 0|1|2,
  "lower_bound_mechanism": "<short name or 'none'>",
  "upper_bound_mechanism": "<short name or 'none'>",
  "wrong_answer_stated": true|false,
  "notes": "<=60 words"
}
```

Field mapping for this problem: "answer" = the target $OM = ON$ is correctly stated as the goal with a correct overall strategy (2 explicit and correct, 1 vague, 0 absent/wrong); "reduction" = (T); "lower_bound" = (K) key circle containing A, K, L proved; "upper_bound" = (F) finish $OM = ON$;

"wrong_answer_stated" = true if the write-up claims a false intermediate lemma or a wrong construction.

The P1 and P5 closure rubrics, used by the replication (Appendix G) and by the planted study’s witness and closure gates, are printed in Appendix B.

Table 8: Evaluator variation: the trajectory corpus of Appendix F (left) and its replication on P1, P4, P5 (right); i / r / c = improved / regressed / first crossing.

Measure	P3, P6, synthetic P2 (§6)	P1, P4, P5 (Appendix G)
identical-text controls: score differs	0 of 15 adjacent pairs (layer 0)	0 of 23 repeated texts (Claude)
same-text repeat disagreement on S	GLM 0/70; Claude 14/504 ($q=0.03$)	Claude 7/246 ($q=0.03$), 3 P5 texts
κ : GLM–Kimi / GLM–Claude / Kimi–Claude	0.81 / 0.78 / 0.78	Claude–GPT 0.79 / 1.00 / 0.82 (P1/P4/P5)
κ : GLM–GPT	0.84 (P3/P6)	Claude–GLM 1.00 / 1.00 / 0.94
conclusions identical, all evaluators (i/r/c)	30 / 29 / 27 of 31 lineages	36 / 35 / 33 of 36 lineages

G Replication on Problems 1, 4, and 5

The evaluator-variation measurements of the trajectory corpus (Appendix F) were repeated on the three problems the paper does not otherwise analyze, with rubrics anchored to the published solutions (closure_rubric_p{1,4,5}.txt; obligation choices in replication_ext/rubric_notes.md), the public campaign’s per-turn tracking-file snapshots plus our retained candidates as texts, and a per-checkpoint pass, three per-text passes, and a second lab as raters (protocol written before the ratings: replication_ext/PROTOCOL.md). Corpus (P1: 38 checkpoints, 30 distinct texts, 12 lineages; P4: 33 checkpoints, 25 distinct texts, 12 lineages; P5: 35 checkpoints, 27 distinct texts, 12 lineages; 106 checkpoints, 82 distinct texts, 36 lineages in all, the nine public runs and our GLM-5.2, GPT-5.6-Sol ultra and Nemotron runs on each problem). Claude Opus 5 rated every checkpoint once and every distinct text three times, GPT-5.6-Sol rated every distinct text once, and GLM-5.3, launched last, had rated 30 of 30 P1 texts, 25 of 25 P4 texts, 27 of 27 P5 texts at the freeze (rubrics anchored to the published solutions; no validity layer). Under one Claude Opus 5 rating per checkpoint, 0 of the 23 texts retained under more than one record received different labels (P1: 0 of 8; P4: 0 of 8; P5: 0 of 7; paper, layer 0: 0 of 14 on P3/P6, 2 of 5 on P2), the same-text disagreement of Claude Opus 5 over three text passes is $q=0.00/0.00/0.09$ on P1/P4/P5 (7 of 246 pairs; 12 of 589 pooling the per-checkpoint ratings; paper: 0.02/0.03 on P3/P6), and Cohen’s κ on S between Claude Opus 5 and GPT-5.6-Sol is 0.79/1.00/0.82 (Claude–GLM-5.3: 1.00, 1.00, 0.94; paper: 0.88/0.95 and 0.94/0.90 on P3/P6). The improved/regressed/first-crossing conclusions are identical under every evaluator with complete coverage for 12/12/11 of 12 on P1, 12/12/12 of 12 on P4, 12/11/10 of 12 on P5 lineages (the five Claude/GPT evaluators are complete on all 36; over the 36 lineages that GLM-5.3 had also rated fully, identical under all six evaluators for 12/12/11 of 12 on P1, 12/12/12 of 12 on P4, 12/11/10 of 12 on P5; paper: P3: 16/16/15 of 17; P6: 12/11/10 of 12); the flips are P1 ours-glm52 (crossing checkpoint 1 under Claude per-checkpoint, Claude pass 1, Claude pass 2, Claude pass 3, Claude majority, GLM; none under GPT); P5 deedy-deepseek-v4-pro (crossing checkpoint 5 under Claude per-checkpoint, Claude pass 1, Claude pass 3, Claude majority, GLM; checkpoint 3 under Claude pass 2; checkpoint 7 under GPT); P5 ours-nemotron (regressed yes under Claude per-checkpoint, Claude pass 3, GPT; no under Claude pass 1, Claude pass 2, GLM; crossing checkpoint 4 under Claude per-checkpoint, Claude pass 1, Claude pass 3, GPT, GLM; checkpoint 3 under Claude pass 2). Across complete single evaluators the number of $S=1$ checkpoints is 4–5 on P1, 0 on P4, 4–7 on P5, and $S=2$ of the final file agrees with the public first-pass grade ≥ 5 for 27 of 27 public finals under Claude Opus 5 and 27 of 27 under GPT-5.6-Sol.

H Verifier Response to Decisive Evidence: Claims and Protocol

Four local claims taken from the audit were shown to four verifiers (GLM-5.3 via its API with thinking and Kimi K3 via its API, both tool-free; Claude Opus 5 via its command-line interface with tools disabled; GPT-5.6-Sol via the Codex command-line interface in a read-only sandbox with web search disabled but not every tool disabled, so “tool-free” is exact for three verifiers and a sandbox statement for the fourth) in three conditions each: the claim alone, with a valid counterexample, and with a deliberately invalid one. The system prompt asks for the claim’s truth and the witness’s

Table 9: Replication of the evaluator-variation measurements on P1, P4, P5, with the paper’s P3/P6 values for reference.

Problem	lin.	ckpt.	texts	differs	Claude q	κ on S		same conclusions impr./regr./cross.	$S=1$ ckpts	$S=2$ iff grade ≥ 5 Claude, GPT
						Cl-GPT	Cl-GLM			
P1	12	38	30	0/8	0.00	0.79	1.00	12/12/11	4-5	9/9, 9/9
P4	12	33	25	0/8	0.00	1.00	1.00	12/12/12	0	9/9, 9/9
P5	12	35	27	0/7	0.09	0.82	0.94	12/11/10	4-7	9/9, 9/9
P3 (paper)	17	57	48	0/8 ^a	0.02 ^b	0.88	0.94	16/16/15	0-2	8/9, 9/9
P6 (paper)	12	42	36	0/6 ^a	0.03 ^b	0.95	0.90	12/11/10	4-7	8/9, 8/9

identical text: texts retained under more than one record whose records received different S from one rating per checkpoint (P1/P4/P5: Claude Opus 5; ^aP3/P6: the paper’s layer 0, GLM-5.3). q : same-text pairwise disagreement on S over three Claude Opus 5 text passes (^bpaper: four Claude ratings per text). κ : Cohen’s κ on S over distinct texts, Claude first text pass vs. GPT-5.6-Sol / GLM-5.3 (P3/P6 recomputed from the released per-text ratings; a superscript gives n where GLM-5.3 had not rated every text at the freeze). Same conclusions: lineages (of the number in column lin.) whose improved / regressed / first-crossing conclusions coincide under every evaluator with complete coverage. $S=1$: range over complete single evaluators of the number of $S=1$ checkpoints. Grade: public final files with ($S=2$) $\Leftrightarrow$ (first-pass grade ≥ 5) under Claude and GPT (P1/P4/P5: first-pass and final grades coincide; P6 (paper): the Kimi K3 final file follows a repair round, first-pass 3, final 7, and against final grades the P6 cells read 9/9, 9/9).

validity as separate JSON fields and forbids deferring to the proposer’s confidence. Every request and response is released under `verifier_evidence/`; three no-verdict calls were repeated once under `verifier_evidence_ext/`. With a four-times larger budget (post-hoc), GLM-5.3 C2-alone: no verdict again (timed out after 30 min); Kimi K3 C2-alone: claim judged correctly; Kimi K3 C2-invalid: claim judged correctly, witness judged correctly.

Table 10: Verifier response to decisive evidence on four checkable claims.

Claim	truth	GLM-5.3	Kimi K3	Claude Opus 5	GPT-5.6-Sol
C1 nonincreasing $\Rightarrow$ stays > 1	false	✓✓✓✓✓	✓✓✓✓✓	✓✓✓✓✓	✓✓✓✓✓
C2 finitely many minimal transversals	false	∅✓✓✓✓	∅✓✓✓∅	✓✓✓✓✓	✓✓✓×✓
C3 prime support is a minimal hitting set	false	✓✓✓✓✓	✓✓✓✓✓	✓✓✓✓✓	✓✓✓✓✓
C4 dyadic subset sums distinct	true	✓-✓✓✓	✓-✓✓✓	✓-✓✓✓	✓-✓✓✓

Per verifier: claim alone (one glyph), with a valid counterexample and with an invalid one (claim + witness glyphs); ✓ judged correctly, × wrongly, ∅ no verdict (output budget exhausted inside reasoning), – not applicable.

You are a careful mathematical verifier. You will be shown a claim and possibly a proposed counterexample (a witness). Assess them separately and independently. Do not defer to the confidence of the proposer. Do not retrieve or recall published solutions; reason from first principles. Return ONLY a JSON object with exactly these fields: {"claim_true": true|false, "claim_reason": "<=60 words", "witness_valid": true|false|null, "witness_reason": "<=60 words or null"}. witness_valid is null when no witness is given; otherwise it is true only if the witness satisfies every hypothesis of the claim and violates its conclusion.

- C1** (*Nemotron 3 Ultra, IMO 2026 P1 draft accepted by two model gates (invalid local inference)*; false) Let $q_1, q_2, q_3, \dots$ be a sequence of positive real numbers with $q_1 > 1$ and $q_{k+1} \leq q_k$ for every k . Then $q_k > 1$ for every k . *Valid counterexample offered*: The sequence $q_1 = 2, q_2 = 1, q_3 = 1, \dots$ satisfies $q_1 > 1$ and is nonincreasing, but $q_2 = 1$ is not greater than 1. *Invalid counterexample offered*: The sequence $q_1 = 2, q_2 = 3, q_3 = 0.5, \dots$ has $q_3 = 0.5 < 1$, so the claim fails.
- C2** (*GPT-5.6-Sol (default effort), public IMO 2026 P6 run graded 1/7; two frontier annotators rated the proof complete*; false) Every family of nonempty finite sets that is pairwise intersecting (any two members share an element) has only finitely many inclusion-minimal finite transversals (finite sets meeting every member of the family). *Valid counterexample offered*: Take distinct symbols $x_1, x_2, \dots$ and $y_1, y_2, \dots$, and let $F_n = \{x_1, \dots, x_n, y_n\}$ for $n \geq 1$. Any two members share x_1 , so the family is pairwise intersecting. For each $k \geq 1$, $T_k = \{y_1, \dots, y_{k-1}, x_k\}$ meets F_n through y_n when $n < k$ and through x_k when $n \geq k$, so it is a finite transversal; removing y_i misses F_i and removing x_k misses F_k , so it is inclusion-minimal. The T_k are pairwise distinct, so there are infinitely many. *Invalid counterexample offered*: Take $F_n = \{x_n\}$ for $n \geq 1$ with distinct symbols x_n . Then each $T_k = \{x_k\}$ is a minimal finite transversal, so there are infinitely many.
- C3** (*Nemotron 3 Ultra, IMO 2026 P6 draft (false key observation recorded by the gate)*; false) Let $a_1 < a_2 < \dots$ be the greedy sequence of IMO 2026 Problem 6 (each a_{n+1} is the least integer greater than a_n sharing a prime factor with every earlier term). Then for every n the set of prime factors of a_{n+1} is an inclusion-minimal hitting set of the family of prime-factor sets of $a_1, \dots, a_n$. *Valid counterexample offered*: Start with $a_1 = 6, a_2 = 8$. The least integer above 8 sharing a factor with both 6 and 8 is 10, so $a_3 = 10$ with prime set $\{2, 5\}$. The family $\{\{2, 3\}, \{2\}\}$ is already hit by $\{2\}$ alone, so $\{2, 5\}$ is a hitting set but not inclusion-minimal. *Invalid counterexample offered*: Start with $a_1 = 6, a_2 = 8, a_3 = 9$. Then the prime set of a_3 is $\{3\}$, which does not hit $\{2\}$, so the claim fails.
- C4** (*IMO 2026 P3, published lower-bound argument (true statement, control)*; true) For the division of a stick into $n + 1$ parts of lengths $1, 2, 4, \dots, 2^n$ (in some unit), all 2^{n+1} subset sums of the parts are pairwise distinct. *Invalid counterexample offered*: For $n = 2$ the parts are 1, 2, 4; the subsets $\{1, 4\}$ and $\{2, 3\}$ both have sum 5, so two subset sums coincide.

I Conclusion Sensitivity Under Evaluator Replay

Reading guide. This appendix asks whether a lineage’s three trajectory conclusions (improved, regressed, first crossing) survive a change of evaluator or a replay of the same evaluator, and whether disagreement between evaluators flags defective texts (protocol: `closure_repeat_PROTOCOL.md`). Two cautions govern every figure. First, the reference is the defect-corrected layer, which reads no lineage as regressed, so a rule that always reports “no regression” reproduces every regression conclusion: a regression-survival figure equal to one says nothing, and only figures below one, spurious regressions under replay, carry information. Second, survival is agreement with that model-derived, partly corrected reference, not accuracy. The reviewed version printed the full survival, evaluator-choice, instability-source and grading-procedure tables and their paragraphs here; they are unchanged in the supplementary bundle (`conclusion_sensitivity/`), and this version keeps the three parts the body relies on: same-text repeat variability, the replay null, and the alarm.

Same-text repeat variability. Same-text disagreement on S : GLM-5.3, 0 of 70 fixed-setting pairs (69 thinking–thinking, 1 fallback–fallback) and 1 of 3 mixed-mode pairs (P6 p6-gpt-5.6-sol-main-t009: no-thinking fallback $S=2$ against thinking $S=1$), hence 1 of 73 for the adaptive thinking-then-fallback workflow as run; Claude Opus 5, 14 of 504 fixed-setting pairs on 84 texts ($q=0.03$). Completed repeats at freeze: GLM-5.3: rep1 only, 48 P3 + 25 P6 texts, 73 repeat ratings; Claude Opus 5: three repeats of 48 P3 + 36 P6 texts, 252 repeat ratings. GLM-5.3’s one disagreement is a change of thinking mode, not fixed-setting stochastic variation, so every lineage covered by its pool has a deterministic pool. Claude Opus 5’s 14 differing pairs fall on 4 of its 84 texts.

Replay null. Under the replay null (independent same-model draws per checkpoint, texts fixed, exact enumeration), the probability of at least one apparent up-transition is zero for every lineage the GLM-5.3 pool covers and for all but 2 of the 26 lineages the Claude Opus 5 pool covers (0.38 for P6 deedy-gpt-5.6-sol and 0.25 for P3 deedy-kimi-k3), a mean of 0.03 over the covered lineages with at least three checkpoints: evaluator variation can masquerade as progress, but under replay it rarely does.

Alarm target, coverage, and pair sensitivity. The alarm is retrospective everywhere except on the planted texts. Disagreement as an alarm: the target is the credited-defective text, one with a documented defect (a correction entry, that is an independent refutation or a recorded fatal invalid step, or a public final file whose released public grade is at most 4 of 7) that at least one evaluator nevertheless rates $S=2$, because a text every evaluator rates $S<2$ is visible to an operator without any alarm. Over the 90 distinct texts of the applicable lineages (48 P3, 36 P6, 6 P2), the four single evaluators plus the GLM-5.3 repeat where present disagree on S for 7 texts: all 4 credited-defective texts (P3 deedy-gpt-5.6-sol-final, P6 p6-gpt-5.6-sol-main-t009, P6 p6-kimi-k3-round2-t007 and P6 p6-deedy-gpt-5.6-sol-final; 3 distinct defects, the two P6 GPT-default texts sharing one false lemma), hence 4 of the 13 documented-defective texts, the 9 misses being never credited (P3 deedy-deepseek-v4-pro-final, deedy-gpt-5.6-sol-max-final, deedy-gpt-5.6-sol-pro-final, deedy-grok-4.5-final and deedy-muse-spark-1.1-final, public grades 1, 1, 2, 2 and 1 of 7, $S=0$ by every evaluator; P6 p6-deedy-deepseek-v4-pro-final, p6-deedy-gpt-5.6-sol-max-final and p6-deedy-grok-4.5-final, public grade 1 of 7 each, $S=0$ by every evaluator; p6-deedy-muse-spark-1.1-final, public grade 2 of 7, $S=1$ by every evaluator), and 3 texts without any record (P3 kimi-k3-round3-t007 and kimi-k3-round4-t014; P6 p6-kimi-k3-round1-t021). These are counts, not rates: the ascertainment is every correction entry plus every public final with a released grade of at most 4 (not a selected evidence set), texts without a record are unexamined, not known negatives, and the field-level rule (some evaluator scores a field 2 where another scores it < 2) flags 7 texts, the same ones. The recommendation to rate with at least two labs was tested above only as the union of four evaluators and a repeat; pair by pair, the credited-defective texts flagged are GLM–Kimi 1 of 4 (3 flags), GLM–Claude 2 of 4 (3 flags), GLM–GPT 4 of 4 (6 flags on 84 texts), Kimi–Claude 3 of 4 (4 flags), Kimi–GPT 3 of 4 (5 flags on 84 texts), and Claude–GPT 2 of 4 (3 flags on 84 texts), while the GLM-5.3 re-rating against its own repeat flags 1 of the 2 credited-defective texts it covers (1 flag on 73 texts). GLM–GPT reaches all 4 credited-defective texts and GLM–Kimi the fewest (1 of 4); three of the 4 triples reach all 4 (every triple containing GPT-5.6-Sol), so which two labs are chosen decides the result, and with 4 target texts the differences between sets are a handful of texts, not rates. The three flags without a record were examined afterwards as group C of Appendix J: a localized concern or better on 3/3 of them under the witness procedure (grader 3/3) and a substantiated defect on 1/3 (grader 1/3).

Table 11: Cross-evaluator disagreement on S as an alarm for credited-defective texts and all documented-defective texts (applicable lineages).

	Problem	texts	defective texts		flags among			
			documented	credited	flagged	credited	documented	no record
	P3	48	6	1	3	1/1	1/6	2/42
	P6	36	7	3	4	3/3	3/7	1/29
	P2 ^s	6	0	0	0	0/0	0/0	0/6
	all	90	13	4	7	4/4	4/13	3/77

S by evaluator										
Documented-defective text	P	record						GLM rep. 1	credited	flag
			GLM	Kimi	Claude	GPT	GLM			
deedy-gpt-5.6-sol-final	P3	public grade 2/7	2	0	2	0	2	yes	yes	
deedy-deepseek-v4-pro-final	P3	public grade 1/7	0	0	0	0	0	no	no	
deedy-gpt-5.6-sol-max-final	P3	public grade 1/7	0	0	0	0	0	no	no	
deedy-gpt-5.6-sol-pro-final	P3	public grade 2/7	0	0	0	0	0	no	no	
deedy-grok-4.5-final	P3	public grade 2/7	0	0	0	0	0	no	no	
deedy-muse-spark-1.1-final	P3	public grade 1/7	0	0	0	0	0	no	no	
p6-deedy-gpt-5.6-sol-final	P6	correction entry; public grade 1/7	2	2	1	1	—	yes	yes	
p6-gpt-5.6-sol-main-t009	P6	correction entry	2	2	1	1	1	yes	yes	
p6-kimi-k3-round2-t007	P6	correction entry	2	2	2	1	—	yes	yes	
p6-deedy-deepseek-v4-pro-final	P6	public grade 1/7	0	0	0	0	0	no	no	
p6-deedy-gpt-5.6-sol-max-final	P6	public grade 1/7	0	0	0	0	0	no	no	
p6-deedy-grok-4.5-final	P6	public grade 1/7	0	0	0	0	0	no	no	
p6-deedy-muse-spark-1.1-final	P6	public grade 2/7	1	1	1	1	1	no	no	

Evaluators: GLM-5.3 re-rating, Kimi K3, Claude Opus 5, GPT-5.6-Sol (P3/P6) and the GLM-5.3 repeat where present (— = not rated). Documented = every correction entry of validity_overrides.json plus every public final file of an applicable lineage whose released public grade is at most 4 of 7 (P3 from the released grader receipt, P6 from the released grade table; every model, any wording): 13 texts. Credited = documented and rated $S=2$ by at least one evaluator (4 texts carrying 3 distinct defects; the two P6 GPT-default texts share one false lemma); the alarm's target, since a text every evaluator rates $S < 2$ is visible without an alarm. The field-level rule (a field scored 2 by one evaluator and < 2 by another) flags 7 texts, the same ones. Counts, not rates: texts without a record are unexamined, not known negatives.

Table 12: Pair sensitivity of the alarm: flags and hits of each evaluator pair, and of the full panel, on the credited-defective and documented-defective texts.

Evaluator set	texts	flagged	flags among		
			credited	documented	no record
GLM–Kimi	90	3	1/4	1/13	2/77
GLM–Claude	90	3	2/4	2/13	1/77
GLM–GPT	84	6	4/4	4/13	2/71
Kimi–Claude	90	4	3/4	3/13	1/77
Kimi–GPT	84	5	3/4	3/13	2/71
Claude–GPT	84	3	2/4	2/13	1/71
GLM–Kimi–Claude	90	5	3/4	3/13	2/77
GLM–Kimi–GPT	90	7	4/4	4/13	3/77
GLM–Claude–GPT	90	6	4/4	4/13	2/77
Kimi–Claude–GPT	90	6	4/4	4/13	2/77
four single evaluators	90	7	4/4	4/13	3/77
four single evaluators + GLM repeat (the rule above)	90	7	4/4	4/13	3/77
GLM re-rating against its repeat	73	1	1/2	1/11	0/62

A text counts for a set when at least two of its evaluators rated it (GPT-5.6-Sol rated the 84 P3/P6 texts, the GLM-5.3 repeat 73 texts, the others all 90), so a set containing GPT reduces to its other members on P2. Credited = flagged credited-defective texts / credited-defective texts the set rated; documented and no record likewise. Counts, not rates: with 4 credited-defective texts the differences between sets are a handful of texts.

J Witness Discovery on Disputed Texts

Protocol. `witness_discovery/PROTOCOL.md` is prespecified and self-timestamped, not registered: its recorded writing time (04:21 UTC, 6 Sept 2026) precedes the first call (04:31 UTC), and Amendment 1 (04:29 UTC, before launch) added the evidence classes, the ordinary-grader baseline and the separate reporting of known-defect recovery, open disputes and controls, but these times are self-authored, and the first commit containing the protocol (07:08 UTC) postdates the first calls. `DEVIATIONS.md` logs everything that happened afterwards, including one transport change for the GLM checker (its first six checks exhausted a 16k-token cap in thinking; the cap was raised and the closure instrument’s no-thinking retry added, and the failed checks redone). **Cases** (20). A, credited-defective (an independent record refutes a credited mechanism and at least one evaluator scored $S=2$): P6 GPT-default turn 9 and final (one shared false lemma), P6 Kimi round-2 turn 7, P3 GPT-default final. B, documented-defective and never credited (every evaluator $S=0$): the P3 finals of Grok 4.5, Muse Spark 1.1 and DeepSeek V4 Pro, three of the nine such texts the systematic ascertainment of Appendix I finds. C, alarm flags without a record: P3 Kimi round-3 turn 7 and round-4 turn 14, P6 Kimi round-1 turn 21. D, replication flips: the P1 GLM-5.2 candidate (GPT $S=1$ against Claude and GLM $S=2$), P5 DeepSeek turns 33 and 38 (crossing index differs by evaluator), the P5 Nemotron candidate whose three Claude passes read 1, 2, 0. E, high-grade agreement controls (public grade 7, unanimous $S=2$): P3 Claude Fable 5 and GPT xhigh finals, P6 GPT pro final, P1 Kimi final, P4 Claude final; their correctness is not established by those grades, and finding no witness does not establish it. F, low control: P6 Grok final. Authorship overlap is declared in the protocol (GPT-5.6-Sol wrote five of the texts, Claude Fable 5 two); proposers saw the problem’s marking scheme and the text; graders saw the coordinator scale, which forbids consulting a solution or a lemma list, and the text (next paragraph).

Procedures and budget. Witness procedure: one call per text and model (GPT-5.6-Sol through the Codex CLI in a read-only sandbox with web search disabled, reasoning effort high; Claude Opus 5 through the Claude Code CLI with no tools) with a system prompt that contains the problem’s closure rubric (the marking scheme), its return block replaced by instructions to return, per obligation, `written_and_valid`, `written_but_invalid` or `not_written`, a verbatim quote of 40–400 characters (matched mechanically to the text) and a witness. Ordinary grader: the same models and transports, one call per text, with the paper’s 0–7 coordinator rubric extended to list up to three errors (description and optional quote). Each arm therefore made 40 first-stage calls (20 texts, two models). The arms are matched only in first-stage call count and in the checking rule. They are not matched in information: the witness prompt contains the problem’s marking scheme (4072–6257 characters across the 5 problems; it lists the steps a complete proof needs, states the answer on P3, P4 and P5 and names example wrong approaches on P1, P4, P5 and P6), while the grader prompt (1575–2064 characters) forbids consulting a published solution or a fixed list of required lemmas. Nor are they matched in instruction (the witness prompt says not to repair the argument for the author; the grade scale places a repairable error at 5–6), in tokens or in time. The reviewed version named only tokens and time as unmatched. The planted study’s witness and grade gates use the same prompts (Appendix L.6). Checking: at most two objections per model, text and procedure (the proposer’s lower/upper witnesses; the grader’s two most severe errors) were judged by the two models that did not produce them (GPT’s by Claude Opus 5 and GLM-5.3; Claude’s by GPT-5.6-Sol and GLM-5.3) with the full text, returning `quote_faithful`, `verdict`, `evidence_kind` and `fatal`: 84 checks in the witness arm and 108 in the grader arm. The checker prompt shows the obligation, the quote, the objection and a `Claimed` status field, which carries the proposer’s status (`written_but_invalid` or `not_written`) for a witness objection and the fixed string “error reported by an ordinary grader” for a grader objection; checkers were therefore not told which model produced an objection but could read which procedure did. The protocol’s statement that they saw neither is wrong, and every comparison of checker behaviour across procedures, in particular the unsupported-allegation counts (0 against 2), is exploratory. Rule: substantiated if both confirm and both classify the evidence as a counterexample or an invalid inference; localized if both confirm otherwise; unsupported if both reject; unresolved otherwise; an objection whose second checker had not returned at the analysis freeze is classified from the returned checker and marked single-checker. The rule uses agreement on the verdict and the evidence kind and, as the protocol declared, ignores the `fatal` field; `quote_faithful` is the checker’s reading of the quote in context and is distinct from mechanical anchoring, which the rule does not enforce. Anchored quotes: 42 of

42 witness-procedure quotes and 50 of 54 grader quotes matched the text verbatim after whitespace and markup normalization.

Results. Table 14 gives every text; Table 13 the counts per group; RESULTS.md prints every quote, witness and checker explanation. Both procedures recovered every documented defect: 6 distinct defects on 7 texts, the two P6 GPT-default texts sharing one false lemma. Proposer status agreement: 55/60 obligation verdicts; delivery failures: 0. Objections checked: witness procedure 42 (19 substantiated, 22 localized, 0 unsupported, 1 unresolved); grader 54 (18, 31, 2). Recovery of a documented defect requires a substantiated or localized objection whose target coincides with the record (recovery.json: target-match judgments produced by a model, Claude in Claude Code, under the author’s direction from the quotes before the checker verdicts were read, and reviewed by the author); the same file records, for each open dispute, which side of the earlier evaluator disagreement the objections support and which lineage conclusion changes if the defect is applied with the criterion of validity_overrides.json. Executable checks for the finite counterexamples (P3 constructions by exact claiming value, the P6 transversal family, the P1 board, order dependence of a P3 chain value, the P5 verification chain) are in scripts/witness_checks.py. The one unresolved witness-procedure objection is GPT’s non-sequitur on the P6 GPT-default final’s branch argument (Claude confirmed, GLM-5.3 rejected; Claude’s own witness on the same step was confirmed by both); the grader’s two unsupported allegations are on P5 texts (a case the text handles elsewhere; an imprecise but valid symmetry step).

Severity. Requiring both checkers to call the error fatal changes the open-dispute counts from 4/7 versus 4/7 to 4/7 versus 2/7; the fatal judgments that differ between procedures on the same quoted passage are 2 of the 8 (text, checker) pairs compared (all on the P5 Nemotron display). Per group (witness procedure / grader), texts with a substantiated defect that both checkers call fatal: A 4/4/3/4, B 1/3/2/3, C 1/3/1/3, D 3/4/1/4, E 0/5/0/5, F 0/1/0/1 (second row of Table 13; results.json, key fatal_layer). Under this condition the grader arm loses P5 DeepSeek turn 38, where its only substantiated objection is Claude’s *4cx/4cy* expansion typo (nonfatal for both checkers), and the P5 Nemotron display; the witness arm loses nothing. On that display GPT’s witness and GPT’s grader error quote the identical formula: Claude Opus 5 and GLM-5.3 confirmed the error under both procedures and called it fatal when it arrived as a witness (written_but_invalid, with a load-bearing explanation) and nonfatal when it arrived as the grader’s description of a repairable typo. The two objection texts differ in wording as well as in the procedure label, so the experiment does not isolate what caused the change; it is an observation that a confirmed local error does not carry a stable judgment of its consequence for the obligation. This layer is post hoc and diagnostic; the classification rule is the prespecified one.

Cost (retained responses; results.json, key cost). Each arm made 40 first-stage calls (20 texts, two models; mean 58.4 s for GPT-5.6-Sol and 72.2 s for Claude Opus 5 in the witness procedure, 43.0 and 105.8 s in the grader), followed by 84 checks in the witness arm and 108 in the grader arm (two checkers per objection). Retained primary-model usage in tokens: witness arm 1,678,756 (831,888 input, 457,530 output, 148,634 cache read, 240,704 cache creation), grader arm 2,054,699 (913,858 input, 677,645 output, 193,557 cache read, 269,639 cache creation). Helper-model usage the Claude CLI records beside the requested model adds 188,681 and 223,172 tokens; GLM-5.3 attempts retried before the retained reply, where their usage was stored, add 113,771 and 56,857. Mean checker call time: GPT-5.6-Sol 22.3 s, Claude Opus 5 32.6 s, GLM-5.3 351.8 s over 90 of 96 timed checks (6 records, all on deedy-deepseek-v4-pro-final, carry no timing metadata). Response files replaced by a rerun are not retained, so these are retained-response totals, not total spend. The untimed records are the smoke-test checks made before the GLM transport change; six early GLM checks exhausted a 16k-token cap in thinking and were redone after the cap was raised to 32k with a 12k thinking budget and the closure instrument’s no-thinking retry (DEVIATIONS.md). The Claude CLI records a small auxiliary Haiku call beside each Opus 5 invocation, so “one call” is one CLI invocation, not one provider inference.

K Reference-Anchored Adjudication: Ledger, Protocol, Results

Ledger. reference_adjudication/LEDGER.md lists, for P1, P3, P5 and P6, the statements every complete proof establishes, produced by a model (Claude, in Claude Code) under the author’s direction from Chen [2026] (version of 3 September 2026), with page references, and reviewed by the author, and the mapping of each v6 dispute to a statement. Cross-check: the AoPS wiki page for P5

Table 13: Witness discovery: texts per group with each outcome, witness procedure / ordinary grader. A credited-defective; B documented-defective, never credited; C alarm flags without record; D replication flips; E high-grade agreement controls; F low control.

	A (n=4)	B (n=3)	C (n=3)	D (n=4)	E (n=5)	F (n=1)
substantiated defect	4 / 3	1 / 2	1 / 1	3 / 3	0 / 0	0 / 0
both checkers fatal	4 / 3	1 / 2	1 / 1	3 / 1	0 / 0	0 / 0
localized concern or better	4 / 4	3 / 3	3 / 3	3 / 3	0 / 0	1 / 1

Table 14: Witness discovery per text: S_{valid} (witness procedure) or grade (ordinary grader) followed by the class of the text’s best objection: **S** substantiated defect, **L** localized concern, **U** unsupported allegation, **?** unresolved, **–** no objection.

Grp	Text	Witness procedure		Ordinary grader	
		GPT	Claude	GPT	Claude
A	p6-gpt-5.6-sol-main-t009	1/S	1/S	3/S	1/S
A	p6-deedy-gpt-5.6-sol-final	1/?	1/S	3/S	1/S
A	p6-kimi-k3-round2-t007	1/S	1/S	4/S	5/L
A	deedy-gpt-5.6-sol-final	0/S	0/S	3/L	3/L
B	deedy-grok-4.5-final	0/S	0/S	2/L	1/L
B	deedy-muse-spark-1.1-final	0/L	0/L	2/L	1/S
B	deedy-deepseek-v4-pro-final	0/L	0/L	1/L	0/S
C	kimi-k3-round3-t007	0/L	0/L	3/L	2/L
C	kimi-k3-round4-t014	1/S	1/S	6/S	7/S
C	p6-kimi-k3-round1-t021	0/L	0/L	2/L	1/L
D	p1-glm52-2ede1c6b-v1	2/–	2/–	7/–	7/–
D	p5-deepseek-v4-pro-main-t033	1/S	1/S	6/S	6/S
D	p5-deepseek-v4-pro-main-t038	1/L	1/S	6/L	6/S
D	p5-nemotron-2587b733-v2	1/S	1/S	6/S	6/?
E	deedy-claude-fable-5-final	2/–	2/–	7/–	7/–
E	deedy-gpt-5.6-sol-xhigh-final	2/–	2/–	7/–	7/–
E	p6-deedy-gpt-5.6-sol-pro-final	2/–	2/–	7/–	7/–
E	p1-deedy-kimi-k3-final	2/–	2/–	7/–	7/–
E	p4-deedy-claude-fable-5-final	2/–	2/–	7/–	7/–
F	p6-deedy-grok-4.5-final	0/L	0/L	1/L	0/L

(three community solutions) agrees with the notes on the answer and on the first three statements; two of its three solutions are invalid as written (one concludes $f(x) + y = f(y) + x$ from two inequalities that share a lower bound; one asserts injectivity without proof), and the single community solution on the P3 page assumes Xiang’s only reply is to halve every larger segment (the false-guarantee failure mode of the model texts) and proves no upper bound, the P6 page’s first solution asserts its finiteness lemma by pigeonhole while its second proves it by a key-terms argument equivalent to the notes’ minimal terms, and the P1 page agrees on both statements; so the wiki confirms which statements are required and is used only where it agrees with the notes (the author saved the P3, P6 and P1 pages by hand because the site blocks automated access; the P5 page was fetched in a browser session). The ledger settles which statements are required and the truth of the claims the disputes turn on; it does not settle the convention for a false display beside a correct argument. A model check of 29 September 2026 (the double-check of Appendix P) found two further ledger issues, neither of which changes a recorded judgment: in `LEDGER_v2.md` the P6 entry L6.2 gives as “equivalently” a condition that is only sufficient for a fixed prime set (no judgment used that edition), and the explicit survivor argument of the P1 entry L1.1 is the ledger’s own gloss, not Chen’s text (`reference_adjudication/ADDENDUM_2026-09-29.md`).

Protocol (`PROTOCOL.md`, prespecified and self-timestamped: its recorded writing time, 14:27 UTC on 6 September 2026, precedes the first call at 14:29 UTC, and the first commit containing it, at 15:09 UTC, follows the first calls; deviations in `DEVIATIONS.md`). Stage 1: each of the 96 v6 objections (quote and objection text only; no procedure label, no claimed status, no proposer identity, no v6 verdicts) was judged by GPT-5.6-Sol and Claude Opus 5 with the closure rubric, the ledger and the full text, returning the affected statement, whether the objection’s local claim is correct, whether the statement is established elsewhere (verbatim quote, anchored mechanically), the status as written (established / not established / false display) and a repair distance; the checker is told that the objection’s own opinion of severity is not evidence. A checker may judge an objection produced

by its own lab: the two checkers here, as in the planted study’s reference gate, are the labs whose models produced the objections, so same-lab and other-lab agreement are reported separately. Stage 2: ledger-guided closure ratings of the 20 texts by GPT-5.6-Sol, Claude Opus 5 and GLM-5.3 (coverage GPT 20/20, Claude 20/20, GLM 20/20). Rules: an objection’s status is the two checkers’ status when they agree, else split; a text’s obligation is not established when an objection on it is adjudicated not established with a named statement; false display only when that is the only non-established outcome.

Aggregation (v8). Per objection: a consensus status when both checkers agree, else a split with both statuses kept (four not-established/false-display, two not-established/established, one established/false-display); the checkers named the same statement on 95 of 96 objections and endorsed the local objection on 95 (same-lab checker 95/96, other-lab 95/96; this is endorsement, not agreement on severity). Per scored obligation, under each convention: not established (a consensus negative objection maps to it), unresolved (a split, or a statement with conflicting consensus judgments), established (a consensus positive objection with an anchored quote), or no objection, never merged with established. Two conventions: as written (AW), a false display counts against the statement; repair-tolerant (RT), it does not. This repair-tolerant convention is broader than the planted study’s, under which only a false line that no later line cites is acceptable (a harmless edit): here a false display beside an argument that does establish the statement is tolerated, which can re-admit a fatal edit (Appendix L.7). “False display” is an adjudication status, not the planted study’s harmless class. *D* counts obligations failed by consensus; splits make it an interval. **Results.** Objections both checked: 96 of 96; same statement named by both 95 (one objection was mapped to two statements and is attributed to both); same status 89 (83 not established, 5 false display, 1 established); the 7 splits are four not-established/false-display, two not-established/established, one established/false-display (GPT/Claude), on 4 texts, the stricter checker being GPT on six of the seven splits and Claude on one. Grouping judgments by checker, text and statement under two or more objections gives 48 groups; statuses varied across objections on the same statement in 8 (6 after collapsing to the repair-tolerant reading): defect content and framing were not held fixed and no objection was repeated verbatim, so this is variability across presentations, not a wording effect. Scored obligations (19 texts; the P4 control has no ledger statements): 38, of which 21 not established as written and 19 repair-tolerant, 1 mixed (consensus positive and consensus negative judgments on the same statement), unresolved 0 as written and 2 repair-tolerant, no objection 16, established by consensus 0. The not-contradicted score is the band $[2 - D - U, 2 - D]$ with *D* the obligations failed by consensus and *U* the unresolved or mixed ones; ledger-guided ratings match its upper end on GPT 17/19, Claude 15/19, GLM 14/19 texts as written and GPT 15/19, Claude 16/19, GLM 15/19 repair-tolerant, and fall inside the band on GPT 17/19, Claude 16/19, GLM 15/19 and GPT 17/19, Claude 17/19, GLM 16/19. Anchoring: established-elsewhere quotes 9/11; positive ledger-statement ratings with an anchored quote 63/96 (GPT 26/32, Claude 20/32, GLM 17/32; the denominator is every “yes” statement rating over the 96 rater-statement pairs). Per text: both DeepSeek P5 texts omit the nonnegative-difference step by consensus (four objections each), so their first $S=2$ crossing cannot be at turn 33 or 38 (turn 39, GPT’s index, was not adjudicated); the P1 candidate drew no objection; Kimi P3 round 4’s reduction step fails by consensus, and its upper bound drew four objections on one step (decreasing order versus the lemma’s ordering): Claude false display on all four, GPT not established on three and false display on the one framed “purely cosmetic”; Nemotron’s sufficiency display is false display twice and split once, and its necessity step is not established, established and split across three objections. Disputed texts: fail a scored obligation as written 6/7, repair-tolerant 4/7 (5/7 counting the mixed one), unresolved repair-tolerant: Kimi P3 round 4 and Nemotron P5. Community solutions on the AoPS pages: eight checked (three P5, one P3, two P6, two P1); four contain invalid inferences as written (two P5, the P3 one, the first P6 one). All stage-1 and stage-2 judgments used ledger v1: every request reproduces the statement list generated from LEDGER.md, which itself also records provenance and cross-checks; LEDGER_v2.md records two literal corrections (L6.1 restricted to $x \geq a_1$; L3.2 requiring two distinct disjoint subsets) and tags each statement as a route-independent requirement or a mechanism of the reference route; no judgment was re-run, and no scored outcome depends on either correction: L6.1 is outside the scored obligations and its only objection concerns a text that supplies no argument for it at all, and none of the 22 L3.2 objections concerns an empty pair. Effects of presenting the corrected ledger to the models are unmeasured. RESULTS.md prints every objection with both checkers’ statements, statuses and explanations; Table 16 lists every objection on the seven disputed texts.

Table 15: Same ledger statement, different presentations of the same defect (statuses as written; every objection on the seven disputed texts, both checkers and repair sizes: Appendix Table 16).

Statement (step)	Objection	Presentation	GPT	Claude
Nemotron P5, sufficiency (middle term $(x+y+2c)/2$)	grader-gpt/e1	“repairable typo”, no example	false disp.	false disp.
	witness-gpt/lower	counterexample $x=y=c=1$	false disp.	false disp.
	witness-claude/lower	$c=1, x=2, y=1$; “does not verify the required chain”	not est.	false disp.
Kimi P3 round 4, upper bound (decreasing vs. lemma’s order)	grader-gpt/e1	“must explicitly process the constructed ordering”	not est.	false disp.
	witness-gpt/upper	$\{10, 9, 8, 7, 5\}$: residue 5 vs. 1	not est.	false disp.
	grader-claude/e0	“purely cosmetic”	false disp.	false disp.
	witness-claude/upper	$\{8, 7, 6, 5, 4\}$: value 4 vs. 0	not est.	false disp.

Table 16: Every objection on the seven disputed texts: ledger statement(s) named, each checker’s status as written, and repair distances.

Text	Objection	Stmt	GPT	Claude	Repair G/C
Kimi P3 round 3	grader-gpt/e0	L3.2	not est.	not est.	lemma/lemma
	grader-gpt/e1	L3.3	not est.	not est.	lemma/lemma
	witness-gpt/lower	L3.3	not est.	not est.	lemma/lemma
	witness-gpt/upper	L3.2	not est.	not est.	lemma/lemma
	grader-claude/e0	L3.2	not est.	not est.	lemma/none
	grader-claude/e1	L3.1	not est.	not est.	lemma/lemma
	witness-claude/lower	L3.3	not est.	not est.	lemma/lemma
	witness-claude/upper	L3.2	not est.	not est.	lemma/lemma
Kimi P3 round 4	grader-gpt/e0	L3.1	not est.	not est.	lemma/sentence
	grader-gpt/e1	L3.2	not est.	false disp.	sentence/sentence
	witness-gpt/upper	L3.2	not est.	false disp.	sentence/sentence
	grader-claude/e0	L3.2	false disp.	false disp.	sentence/sentence
	witness-claude/upper	L3.2	not est.	false disp.	sentence/sentence
Kimi P6 round 1	grader-gpt/e0	L6.2	not est.	not est.	lemma/lemma
	grader-gpt/e1	L6.1	not est.	not est.	lemma/lemma
	witness-gpt/lower	L6.2	not est.	not est.	lemma/lemma
	witness-gpt/upper	L6.3	not est.	not est.	lemma/lemma
	grader-claude/e0	L6.2	not est.	not est.	lemma/lemma
	grader-claude/e1	L6.2	not est.	not est.	lemma/lemma
	witness-claude/lower	L6.2	not est.	not est.	lemma/lemma
	witness-claude/upper	L6.3	not est.	not est.	lemma/lemma
GLM P1 candidate	(no objection)	–	–	–	–
DeepSeek P5 t033	grader-gpt/e0	L5.3	not est.	not est.	sentence/sentence
	grader-gpt/e1	L5.5	not est.	est.	sentence/none
	witness-gpt/upper	L5.3	not est.	not est.	sentence/sentence
	grader-claude/e0	L5.3	not est.	not est.	sentence/sentence
	grader-claude/e1	L5.5	false disp.	false disp.	symbol/symbol
	witness-claude/upper	L5.3	not est.	not est.	sentence/sentence
DeepSeek P5 t038	grader-gpt/e0	L5.3	not est.	not est.	sentence/sentence
	grader-gpt/e1	L5.5	not est.	est.	sentence/sentence
	witness-gpt/upper	L5.3	not est.	not est.	sentence/sentence
	grader-claude/e0	L5.3	not est.	not est.	sentence/sentence
	grader-claude/e1	L5.5	false disp.	false disp.	symbol/symbol
	witness-claude/upper	L5.3	not est.	not est.	sentence/sentence
Nemotron P5	grader-gpt/e0	L5.4	not est.	not est.	sentence/sentence
	grader-gpt/e1	L5.1	false disp.	false disp.	symbol/symbol
	witness-gpt/lower	L5.1	false disp.	false disp.	symbol/symbol
	grader-claude/e0	L5.4	est.	false disp.	none/sentence
	grader-claude/e1	L5.4	est.	est.	none/none
	witness-claude/lower	L5.1	not est.	false disp.	symbol/symbol

Table 17: Reference-anchored outcome per text and scored obligation under the as-written (AW) and repair-tolerant (RT) conventions: $\times$ not established by consensus, $?$ unresolved (split), m mixed, $\checkmark$ established by consensus, $-$ no objection; D = obligations failed; ledger-guided ratings S_{ref} by GPT / Claude / GLM; $v6 S_{\text{valid}}$ and grades.

Grp	Text	AW		RT		D		ratings	$v6 S_{\text{valid}}$	grades
		lo	up	lo	up	AW	RT	G/C/GLM	G/C	G/C
A	p6-gpt-5.6-sol-main-t009	$\times$	$-$	$\times$	$-$	1	1	1/0/0	1/1	3/1
A	p6-deedy-gpt-5.6-sol-final	$\times$	$-$	$\times$	$-$	1	1	0/0/0	1/1	3/1
A	p6-kimi-k3-round2-t007	$\times$	$-$	$\times$	$-$	1	1	1/1/2	1/1	4/5
A	deedy-gpt-5.6-sol-final	$\times$	$\times$	$\times$	$\times$	2	2	0/0/0	0/0	3/3
B	deedy-grok-4.5-final	$\times$	$\times$	$\times$	$\times$	2	2	0/0/0	0/0	2/1
B	deedy-muse-spark-1.1-final	$\times$	$\times$	$\times$	$\times$	2	2	0/0/0	0/0	2/1
B	deedy-deepseek-v4-pro-final	$\times$	$\times$	$\times$	$\times$	2	2	0/0/0	0/0	1/0
C	kimi-k3-round3-t007	$\times$	$\times$	$\times$	$\times$	2	2	1/0/0	0/0	3/2
C	kimi-k3-round4-t014	$-$	$\times$	$-$	$?$	1	0	1/2/2	1/1	6/7
C	p6-kimi-k3-round1-t021	$\times$	$\times$	$\times$	$\times$	2	2	0/0/0	0/0	2/1
D	p1-glm52-2ede1c6b-v1	$-$	$-$	$-$	$-$	0	0	2/2/2	2/2	7/7
D	p5-deepseek-v4-pro-main-t033	$-$	$\times$	$-$	$\times$	1	1	1/1/1	1/1	6/6
D	p5-deepseek-v4-pro-main-t038	$-$	$\times$	$-$	$\times$	1	1	1/1/1	1/1	6/6
D	p5-nemotron-2587b733-v2	$\times$	m	$?$	m	1	0	1/0/0	1/1	6/6
E	deedy-claude-fable-5-final	$-$	$-$	$-$	$-$	0	0	2/2/2	2/2	7/7
E	deedy-gpt-5.6-sol-xhigh-final	$-$	$-$	$-$	$-$	0	0	2/2/2	2/2	7/7
E	p6-deedy-gpt-5.6-sol-pro-final	$-$	$-$	$-$	$-$	0	0	2/2/2	2/2	7/7
E	p1-deedy-kimi-k3-final	$-$	$-$	$-$	$-$	0	0	2/2/2	2/2	7/7
E	p4-deedy-claude-fable-5-final	n/a	n/a	n/a	n/a	n/a	n/a	2/2/2	2/2	7/7
F	p6-deedy-grok-4.5-final	$\times$	$\times$	$\times$	$\times$	2	2	0/0/0	0/0	1/0

L Planted Defects: Acceptance Gates Against Constructed Ground Truth

Figures are from the first run of the grade gate (P, 6 September 2026) unless the same-day re-query (R) is named; the witness, closure and reference gates were run once in this paper’s protocol (Appendix S reports a later re-run of the witness gate). Broken means a fatal or gap edit; sound means an original or a harmless edit, sound only under the repair-tolerant convention.

L.1 Protocol, provenance and the status of each read

Who made the texts. The study design is the author’s, with design options drafted with AI assistants at the author’s request; AI agents implemented it, and the author wrote none of the edits. A coordinating agent in Claude Code (Claude Fable 5.1) selected the eight base proofs, wrote the specification of the three edit kinds and their intended labels (PROTOCOL_PLANTINGS.md), and suggested candidate edits for each problem, among them the false monotone-quantity lemma of both P1 fatal edits and, word for word, the false finite-transversal lemma of the fatal edit on the public GPT-5.6-Sol pro P6 proof. Three Claude Fable 5.1 subagents, one per problem group, wrote every edit, every dependency argument, every intended label and all 16 executable checks (6 September 2026, 19:05–19:13 UTC). The author decided the design (three edit kinds per proof, each published as a diff with a dependency argument and a check, with the assistant doing the edits) before any edit existed, choosing to run it against the assistant’s recommendation, and did not read or approve individual edits before the gates ran. Before launch only the assistant examined the edits: every check passed, and it spot-checked four of the 24 diffs. The first examination by anyone else was an internal model-run review that the author requested after the analysis freeze. The reviewed version’s authorship wording is corrected accordingly (Appendix Q, row Q8); “we” in this paper means the study as a whole.

Protocol status. PROTOCOL_PLANTINGS.md is prespecified and self-timestamped, not registered. It was written before the mutant calls and committed at launch, 20 s after the earliest recorded mutant request, and neither it nor plantings.json has been edited since; every mutant request stored from the launch day, 6 September 2026 (grade, re-query, witness, closure and reference checks: every such file in planted_defects/runs/), carries the text digest recorded for that mutant in plantings.json (412 of 412). Smoke-test screenings of one original text (3 requests) preceded the protocol. Operational notes are in DEVIATIONS.md; the dated addendum planted_defects/ADDENDUM_2026-09-29.md corrects its wording where this appendix does.

Status of each read. Prespecified in the protocol: the gates grade ≥ 5 (the pass mark), closure, witness and witness-plus-reference; false accepts on the 16 fatal and gap texts and false rejects on the 8 originals, per gate; harmless texts accepted per gate and convention; the quote locus; the certificate size; and a separate report for the GPT-authored bases. Added afterwards: the strict mark (grade = 7), the completeness flag, the pass-mark sweep, the 16-text sound set (originals plus harmless edits), the fatal/gap split of the broken count, the area under the curve, the per-lab convention reversal and the routing reading, all read post hoc from the same stored responses with no new calls, and the re-query of the grade gate (runs/grade_rep2/, 22:03–22:22 UTC, after the analysis freeze), which the v10 internal review did not request: a Claude Code subagent running Claude Sonnet 5 suggested it, Claude Opus 5 recommended it and the author approved it. “Prespecified (self-timestamped)” replaces the reviewed version’s registration wording (row Q6).

Gates. The pass mark, the strict mark and the completeness flag are read from one grading response; the completeness flag and the strict mark were added after the internal model-run reviews, with no additional calls. The closure gate requires $S=2$; the witness gate requires $S_{\text{valid}}=2$, where S_{valid} counts the two scored fields of the marking scheme, lower_bound and upper_bound (the scheme’s own obligations, which are a lower and an upper bound only on P3; reduction is not scored), returned as written and valid (Appendix B); the witness-plus-reference gate adjudicates witness objections against the ledger (accept iff no scored obligation has a consensus not-established objection and no split; the as-written convention also rejects on a consensus false display). GPT-5.6-Sol and Claude Opus 5 ran every gate on all 24 mutants and 8 originals, GLM-5.3 the closure gate (32/32 texts returned before the freeze); a combined gate accepts only if every model accepts. The prespecified comparator for the reference gate is the witness-only variant; the pooled variant, which adds the graders’ own error lists, is reported for completeness. “The reviews” and “a reviewer” in the reviewed version and in DEVIATIONS.md are these internal pre-submission reviews run by language models (Appendix P), not the workshop’s referees.

L.2 Bases

The eight base texts were accepted at selection (Table 18): four public texts carry the public campaign’s first-pass verifier grade 7 (Claude-based verifier agents, not the IMO jury) and unanimous model closure ratings $S=2$ (P1 Kimi K3, P3 Claude Fable 5 and GPT-5.6-Sol xhigh, P6 GPT-5.6-Sol pro), and four GPT-5.6-Sol ultra candidates from the author’s campaign (P1, P3, P5, P6) passed GPT-5.6-Sol’s own review-and-verify gate and unanimous model closure ratings. “Accepted” means those model verdicts and nothing more; for five bases the accepting verdict came from the lab that wrote the base. GPT-5.6-Sol wrote six of the eight, and the bases cover four of the 6 problems. The coordinating agent chose them among texts every model evaluator accepted, so they are not a sample; a base’s four texts share most of their content, so the 32 texts are not independent observations, and we report counts, not rates.

The author compared the six GPT-5.6-Sol proofs of the author’s campaign, four of which are bases here, with Chen [2026] and found them consistent; this was not a line-by-line or expert check, and no written record of it was kept. The reviewed version presented this comparison as a human check against the published solutions (row Q13 quotes it). On 5 September 2026 a Claude model also compared those six proofs with the notes step by step, and drafts before the reviewed version described that model comparison as a human line-by-line read. On 29 September 2026 a model double-check (Claude Fable 5.1 agents, each followed by an adversarial verifier; a model check, not a human or expert review) compared the six GPT-5.6-Sol bases of this study, the author’s four and the two public ones, with the notes (version of 27 September 2026) and found all of them consistent; the public P6 pro base has one minor scope slip (an interval count stated without the condition $x \geq a_1 - 1$, and applied only where it holds) that nothing depends on.

The Kimi K3 P1 original already carries an unused false closing remark (its characterization of the answer fails for primes that divide no a_i). GPT-5.6-Sol grades it 6, and four decision variants reject it: the strict mark with GPT-5.6-Sol alone or with both graders required, and the two as-written reference variants that use the graders’ objections. Relabelling it reject under the as-written convention leaves the convention reversal of the first run intact (Appendix L.7).

Table 18: The eight bases. Accepted by: the public campaign’s first-pass verifier grade (Claude-based verifier agents, not the IMO jury) or the author’s campaign’s own GPT-5.6-Sol review-and-verify gate; every base also has unanimous model closure ratings $S=2$. Same lab: the accepting verdict came from the lab that wrote the base. Last column: each grader’s grade of the original, first run / re-query.

Base	Problem	Written by	Campaign	Accepted by	Same lab	GPT / Claude on it
p1-deedy-kimi-k3-final	P1	Kimi K3	public	public verifier: 7	no	6/6 / 7/7
p1-gpt56-5aca5e52-v1	P1	GPT-5.6-Sol	author’s	own GPT gate	yes	7/7 / 7/7
deedy-claude-fable-5-final	P3	Claude Fable 5	public	public verifier: 7	yes	7/7 / 7/7
deedy-gpt-5.6-sol-xhigh-final	P3	GPT-5.6-Sol	public	public verifier: 7	no	7/7 / 7/7
gpt56-ours-accepted	P3	GPT-5.6-Sol	author’s	own GPT gate	yes	7/7 / 7/7
p5-gpt56-7324ef70-v1	P5	GPT-5.6-Sol	author’s	own GPT gate	yes	7/7 / 7/7
p6-deedy-gpt-5.6-sol-pro-final	P6	GPT-5.6-Sol	public	public verifier: 7	no	7/7 / 7/7
p6-gpt56-ours	P6	GPT-5.6-Sol	author’s	own GPT gate	yes	7/7 / 7/7

L.3 Classes, checks and what they establish

Each base received three single-edit mutations, one per class, generated by `make_mutants.py` from search/replace specifications and published as diffs with their hashes and dependency arguments (`plantings.json`): *fatal* (F: a statement a later line cites is made false, the consumer kept), *gap* (G: the proof of a load-bearing step is deleted and replaced by a one-line assertion or a brief sketch, its statement kept), and *harmless* (H: a false line nothing later uses, usually a display the next line silently corrects; one is an unused worked example). A fatal edit leaves the argument invalid as written, not necessarily the theorem false. “Single-edit” replaces the reviewed wording (row Q4): each mutant is one search/replace edit, but the published diff of p6-gpt56-ours-F shows 2 adjacent hunks. This gives 24 mutants and 32 texts. Intended labels, by construction: fatal and gap edits reject under both conventions, harmless edits reject as written and accept repair-tolerant, and originals accept. So broken means fatal or gap (16 texts), and sound means an original or a harmless edit (16 texts), sound only under the repair-tolerant convention.

What the checks establish. Every fatal and harmless edit has an executable check (planted_checks.py; 16 of 16 pass), and the 8 gap edits have none. A check certifies that the planted statement is false (for a harmless edit, also that the next line continues from the correct value). It does not show that a later line fails, and the dependency arguments are prose; at least three fatal checks confirm, in their own instance, that the consumer’s conclusion still holds. P5 release note (the frozen check and dependency argument are unchanged): the check refutes the displayed expansion of p5-gpt56-7324ef70-v1-F as an algebraic identity (the correct expansion minus the planted one is $-8(b - c)(c + t)$); no admissible instance falsifies the displayed equality, since every solution has $b = c$; the planted error is an invalid step, and the quoted numeric point, also in the dependency argument, lies outside the hypotheses.

Announced deletions. Every gap replacement contains the words “(standard; details omitted)” (8 of the 8 gap texts, 0 of the other 24 texts), so every deletion announces itself, although the marker does not always end the replacement; unannounced deletions were not tested. The planted edits are presumably easier than natural defects: each is one edit, and every gap edit announces its omission.

Re-planted natural defects and who saw what. 4 edits re-plant defects that the paper describes in natural proofs: both P1 fatal edits plant the false lemma of the Nemotron P1 draft (a quantity that starts above one and never increases stays above one, which the sequence 2, 1 refutes), the fatal edit on the public GPT-5.6-Sol pro P6 proof plants the false finite-transversal lemma of the public GPT-5.6-Sol default P6 run, and the harmless edit on the P5 base plants the $(x + y + 2C)/2$ middle term of the Nemotron P5 display. The P1 closure rubric, written earlier the same day by an AI subagent, names the P1 lemma as an example of a wrong approach, and the P1/P5 planting subagent used it knowingly (plantings.json records “named in the rubric as a wrong approach”). The witness and closure gates saw that rubric; the reference checkers saw the ledger, which calls the P6 transversal lemma false and pre-classifies the $(x + y + 2c)/2$ display as a harmless false line; the grade gate saw neither.

L.4 The five disputed labels

Five intended labels, three fatal and two gap, are disputed: all four P1 fatal-or-gap edits and the P5 sign flip. They were recorded in DEVIATIONS.md after internal model-run reviews before submission, and Table 19 names them with each rule’s decision. The labels and every count computed from them are unchanged: exclusions are sensitivities, and excluding them does not certify the labels that remain. The P5 sign flip p5-gpt56-7324ef70-v1-F, the reviewed version’s only worked fatal example, is disputed for a sharper reason than a one-line repair: every solution has $b = c$, so the planted equality holds at every admissible instance, and the next line states the correct-sign bound; it is an invalid step that the next line silently corrects, the harmless shape. Excluding the disputed labels in the order they were recorded, the pass mark with both graders required accepts 13, 10, 9 and 8 of 16, 13, 12 and 11 broken texts, identically in both runs; the strict mark accepts 1 and then 0 in both runs; the completeness flag accepts 1, 0, 0 and 0 in the first run and 2, 1, 0 and 0 in the re-query; and the witness gate, run once, accepts 0 throughout. After the five exclusions the broken set covers 3 problems. No exclusion figure is given for the closure gate, which accepts an undisputed gap edit.

Stated against the cite rule, a fatal edit is fatal because a later line cites the falsified statement, not because no reader could repair it; the disputes instead ask whether a reader would repair the step in one line. By that criterion the undisputed fatal edits share the disputed feature too: in the first run at least one grader’s response uses the word “repair” on 5 of the 5 undisputed fatal edits and both graders’ on 4 (in the re-query 5 and 5; a keyword read, not a severity coding), p6-gpt56-ours-F of Figure 1 among them, whose inserted line restates a hypothesis that the proof’s own closing section refutes. The narrower description of the dated addendum, a false statement whose needed special case is true, fits two of them: p6-deedy-gpt-5.6-sol-pro-final-F, whose needed bounded special case is true, and deedy-claude-fable-5-final-F, whose check records that the lemma’s conclusion still holds while its written argument breaks; it does not fit p6-gpt56-ours-F, whose check shows that the edited proof’s own conclusion fails. None of them is added as a disputed label, and no count excludes them.

L.5 Prompts

The grading prompt’s first line and instruction block, the witness gate’s instruction block and the P1 and P5 closure rubrics are printed verbatim in Appendix B; the P3 and P6 rubrics are in Appendix F.

Table 19: The five disputed labels (DEVIATIONS.md), with each rule’s decision, both graders required: pass mark, strict mark and completeness flag as first run / re-query; the witness gate ran once. ✓ accept, × reject. “post-v12” and “post-v15” name internal model-run reviews before submission.

Text	Class	Reason recorded	Recorded	Pass	Strict	Flag	Witness
p1-gpt56-5aca5e52-v1-G	G	an exposition deletion rather than a gap	2026-09-06	✓/✓	✓/✓	✓/✓	×
p1-deedy-kimi-k3-final-F	F	false as written but repaired by one line	2026-09-06	✓/✓	×/×	×/×	×
p5-gpt56-7324ef70-v1-F	F	false as written but repaired by one line	2026-09-06	✓/✓	×/×	×/×	×
p1-deedy-kimi-k3-final-G	G	the same kind of exposition omission	2026-09-07 (post-v12)	✓/✓	×/×	×/✓	×
p1-gpt56-5aca5e52-v1-F	F	the same one-line repair of the monotonicity step	2026-09-07 (post-v15)	✓/✓	×/×	×/×	×

The grading prompt’s scale reads “7 = complete and correct proof; 5-6 = essentially complete, with minor gaps or a small error that is straightforwardly repairable”, and it names the complete field without defining it.

L.6 What each arm is given

The reviewed version’s description of the closure and witness gates is corrected here (row Q14). Table 20 gives, per problem, what each arm’s system prompt contains. The grade arm is reference-free: its prompt carries no solution material and asks the grader not to consult or recall one, which cannot remove what a model knows. The witness and closure arms are reference-guided: their prompts carry the problem’s solution-derived marking scheme, with the answer given on P3 and P5, the value of M in P1’s field mapping and the form of L on P6. Every comparison between the witness or closure gate and a rule read from the grade response is therefore reference-guided against reference-free, not information-matched, and the witness gate’s relative outcome cannot be attributed to obligation-by-obligation checking alone.

On P1 the scheme meets the planted edits twice. First, both P1 fatal edits plant the false principle that the P1 rubric names as a wrong approach (near-verbatim on the Kimi base, reworded on the GPT base). On those two texts the strict mark and the completeness flag, read from graders that never saw the scheme, also reject in both runs, so this collision bears only on the comparison between the witness gate and the pass mark, which accepts both texts. The keyed closure gate’s GPT-5.6-Sol rater accepted p1-deedy-kimi-k3-final-F despite the named lemma (the closure pair rejects it), and the P1 scheme also gives the correct replacement argument for the survivor step. Second, the P1 rubric requires the Euclidean step $\gcd(x, y - x) = \gcd(x, y)$ with the cases $x = y$ and zero valuations handled, and both P1 gap edits delete exactly that step. This collision is the witness gate’s whole broken-text margin over the strict mark and the flag: the one broken text that the strict mark and the flag accept in the first run is p1-gpt56-5aca5e52-v1-G, and in the re-query the flag accepts both P1 gap edits. Both collisions fall on the four P1 broken texts, all of them disputed, so the comparisons without the disputed labels contain no P1 text; the scheme’s other asymmetries (answers on P3 and P5, named wrong approaches) still apply there.

Table 20: What each arm’s system prompt contains, per problem (stored request.json; identical across graders and texts of a problem). Characters: system-prompt length. Answer given: “value of M ” and “form of L ” are given in the field mappings of P1 and P6. Solution ban: the grade prompt’s “Do not consult, recall, or compare against any published solution, and do not use a fixed list of required lemmas”.

Problem	Arm	Characters	Required steps	Answer given	Wrong approaches named	Solution ban	“Do not repair”
P1	grade	2039	no	no	no	yes	no
P1	witness	5692	yes	value of M	yes	no	yes
P1	closure	3365	yes	value of M	yes	no	no
P3	grade	1957	no	no	no	yes	no
P3	witness	4072	yes	yes	no	no	yes
P3	closure	1745	yes	yes	no	no	no
P5	grade	1575	no	no	no	yes	no
P5	witness	5519	yes	yes	yes	no	yes
P5	closure	3192	yes	yes	yes	no	no
P6	grade	1807	no	no	no	yes	no
P6	witness	4591	yes	form of L	yes	no	yes
P6	closure	2264	yes	form of L	yes	no	no

L.7 Full results

Prespecified gates (Table 24; RESULTS.md lists every decision and objection). Fatal-or-gap (broken) texts accepted: grade 13/16 (GPT 15/16, Claude 13/16; fatal 8/8, gap 5/8); closure 2/16 for GPT and Claude combined (GPT 3/16, Claude 8/16, GLM 8/16: the closure rating measures written mechanisms, and a false statement whose consumer is kept still reads as written); witness 0/16; witness plus reference 2/16 repair-tolerant and 0/16 as written. Originals rejected: grade 0/8, closure 0/8, witness 0/8, reference 0/8 (the pooled as-written variant rejects 1/8, the Kimi K3 P1 original). Harmless edits accepted: grade 8/8, closure 6/8, witness 3/8 (GPT 3/8, Claude 7/8), so the witness gate refuses 5/8 of the harmless edits; reference 8/8 repair-tolerant and 3/8 as written. By base author, the prespecified separate report: on the six GPT-written bases the grade with both graders required accepted 10/12 of the broken texts and the witness gate 0/12; on the other two, 3/4 and 0/4; per grader and rule, for both runs, in Table 22. GPT-5.6-Sol’s own pass mark accepts 11 of the 12 broken texts on the bases it wrote and 4 of 4 on the others; two other-authored bases cannot separate self-preference, which is documented in general [Panickssery et al., 2024]. Fatal texts were graded 5 to 7 by both models; of the fatal-or-gap texts the grade accepted, 13/13 had the planted hunk quoted in at least one grader’s error list. Objections whose quote falls inside the planted hunk: 78/98 (witness 36/41, grader 42/57). Certificate size against text length (medians over objections, so a text with several objections weighs more than once): a median objection of 631 characters for the witness gate (one run), on texts of median length 8,596, and of 351 for the grader in the first run, on texts of median length 8,717; the reviewed version set both against the first text median.

Under the reading fixed in advance, the witness gate (one run) is “positive”: it has fewer false accepts than the grade at pass mark 5 and rejects no more originals (0 of 8, as the pass mark), the protocol’s false-reject endpoint. Counting the harmless edits as sound, as Table 1 does, it rejects 5 of the 16 sound texts against the pass mark’s 0 (the witness gate was run once): a trade, not a dominance. The repair-tolerant reference convention re-admits two fatal texts (2/16), both disputed labels (p1-deedy-kimi-k3-final-F and p5-gpt56-7324ef70-v1-F), whose false statements the checkers judged repairable beside a correct continuation: the convention dependence of Appendix K, measured against the intended truth.

Decision-rule analysis (added after the v12 internal model-run review; post hoc, offline, from the same stored responses). Table 1 reports the pass mark, the strict mark and the completeness flag; Table 23 adds the closure and witness gates and the routing columns.

Threshold sweep. The pass mark is swept over the whole scale with both graders required, counting acceptances among the 16 broken texts and rejections among the 16 sound ones (Table 21). The cutoff trades broken texts accepted for sound texts rejected, and no cutoff on this rubric keeps both low: in the first run a pass mark of 5 gives (13, 0) and 7 gives (1, 9), and in the re-query (13, 0) and (1, 8). The rubric places a small repairable error at 5–6, so the sweep bounds what a threshold achieves within this rubric, not across rubrics: the 5–6 band explicitly permits straightforward repair, so a pass at 5 is not a certificate that a proof is complete as written, and the mismatch is between a partial-credit scale and its use as a full-proof acceptance gate.

Strict mark and completeness flag. The prompt names the complete field without defining it, while its scale calls 7 “complete” and 5–6 “essentially complete”. In both runs the field is nested between the cuts at 6 and 7: in the first run all 28 responses graded 7 set it and none of the 8 graded 5 or below does, and it departs from “grade = 7” on 4 of 64 responses in the first run and 2 in the re-query, all of them GPT-5.6-Sol responses graded 6. As rules with both graders required, in the first run the strict mark accepts 1 broken text and rejects 9 sound ones and the flag accepts 1 and rejects 6, a difference of 3 sound texts on 3 bases, all in the flag’s favour; with 3 discordant texts, an exact two-sided McNemar test cannot fall below $p = 0.25$ (one-sided: 0.125). In the re-query the strict mark gives (1, 8) and the flag (2, 7), one discordant text each way, so in the re-query neither dominates. In the first run the strict mark’s sound rejections are 8 harmless edits and 1 original (the Kimi K3 P1 text with its own false remark), and the flag’s are all harmless. The one broken text both rules accept in the first run is the disputed p1-gpt56-5aca5e52-v1-G; in the re-query the flag’s 2 broken acceptances are both disputed P1 deletions.

GPT-5.6-Sol’s flat 6 and the area under the curve. In the first run GPT-5.6-Sol graded all 16 fatal and harmless edits 6, so its scores had no variance across the two classes; in the re-query 1 harmless edit received a 7 (15 of the 16 at 6). This is not blindness: in the first run it quoted the harmless edit in 8 of 8 and, with each base as its own control, graded it one point below its own grade of the

original on 7 of 8 bases (6 in the re-query), pricing a false line the same whether or not anything used it. Claude Opus 5 graded 8 of the 8 harmless edits 7 in the first run (7 in the re-query). The flat 6 also holds in the first run on the two bases GPT-5.6-Sol did not write (4 of 4 fatal and harmless edits). The area under the curve for separating fatal from harmless edits by score, ties counted one half, is 0.50 for GPT-5.6-Sol and 0.94 for Claude Opus 5 in the first run (0.56 and 0.88 in the re-query); for GPT-5.6-Sol, whose grades on these edits are all 6 or 7 in both runs, it equals one half plus half the difference between the shares of harmless and fatal edits graded 7, so it measures the repair-tolerant convention, not discrimination; for Claude Opus 5, which also graded a fatal edit 5 in each run, that identity holds only approximately.

Quotes and incomplete marks. On the 13 broken texts that the pass mark accepted with both graders required (in both runs), at least one grader quoted inside the edited passage on 13 (13 in the re-query), and Claude Opus 5 on 13; GPT-5.6-Sol did so on 7 by exact anchoring and on 9 by tolerant anchoring, the two extra quotes differing from the edit only in their math delimiters. In both runs GPT-5.6-Sol's main gap named the planted error on 8 of the 8 fatal edits and graded 8 of them 6. On the 5 accepted gap edits GPT-5.6-Sol's main gap reads "none" on 4, and in the first run its notes call 2 of the omissions routine or adequately sketched (3 in the re-query); the 3 refused gap edits are those at least one grader scored below 5; in both runs they are the P6 deletions and the deletion in the Claude-written P3 base, and no P1 or P5 deletion was refused (per-mutant table, App. R), a post-hoc pattern confounded with problem, and no mechanism is claimed for the difference. In the first run at least one grader marked the text incomplete on 12 of the 13 accepted broken texts and both did on 8; of the 26 responses on them, 5 were a 7 in the first run and 6 in the re-query.

Repair language. The grading prompt's scale defines 5–6 as "essentially complete, with minor gaps or a small error that is straightforwardly repairable". In the first run, repair language ("repairable", "readily repaired", "easily fixed", "minor", "cosmetic", "does not affect") appears in the main gap or error descriptions of 14 of the 32 grading responses on the broken texts, covering 12 of the 16 texts, and in 21 responses on 15 texts counting the free notes field. It is a keyword match, not a severity coding. Split by where the matching objection points, 10 responses quote inside the edited passage, 3 quote a different step, and 1 carries the language only in the unanchored summary; position is not attribution.

Routing. Read as accept on agreement, reject on agreement and refer on disagreement, each rule sends the texts its two graders dispute to a person; a referral is a count of texts, not timed, and what a referral takes was not measured. In the first run the pass mark refers 2 of the 32 texts, passing 13 broken texts and refusing 0 sound ones (identical in the re-query); the flag refers 10, passing 1 and refusing 0 (in the re-query 9, 2 and 1); the strict mark refers 12, passing 1 and refusing 0; and the witness rule, run once, refers 4, passing 0 and refusing 1. In the first run the flag's referrals are 6 harmless edits, 3 announced deletions and 1 fatal edit (the disputed sign error), and the witness rule's are 4 harmless edits. The witness rule's one refused sound text is the planted copy of the Nemotron P5 display (p5-gpt56-7324ef70-v1-H), sound only under the repair-tolerant convention and correctly refused as written. On broken texts the witness rule's gains over the flag (one run) are 4 gap edits that announce the omission, which its instructions class as not written, and 1 disputed sign error; the rest of its fewer referrals are harmless edits. The witness rule is a comparator for the rules read from one grading response, not a proposed rule.

Witness plus reference. The nearest in-paper analogue of reference-guided grading is the witness-plus-reference gate (a ledger of required statements written from a published solution, two checkers): repair-tolerant, in its one run it accepts 2 of the 16 broken texts, both disputed labels (2), and rejects 0 of the 16 sound ones; the as-written variant accepts 0 broken texts and rejects 5 sound ones, all harmless edits; and the pooled variant, which adds the graders' own error lists, accepts 1 and rejects 1. It is not a reimplementing of published reference-guided workflows, and its objections come from the witness prompt, which carried the marking scheme.

Convention reversal. Counting each rule's disagreements with the intended truth per lab over the 32 texts, which lab has fewer errors reverses between the conventions in the first run under the strict mark, the flag and the witness gate, and in the re-query under the strict mark and the flag. Under the flag the GPT-5.6-Sol/Claude Opus 5 errors are RT 10/2 and AW 6/10 in the first run and RT 11/4 and AW 5/10 in the re-query; the strict-mark and witness-gate counts are in §4. With the Kimi K3 P1 original relabelled reject as written, the as-written errors (GPT-5.6-Sol/Claude Opus 5) in the first run become 3/11 for the strict mark, 7/11 for the flag and 4/8 for the witness gate (one run), against

12/2, 10/2 and 5/1 repair-tolerant, so in the first run the reversal survives the relabelling. With the five disputed labels set aside (27 texts), the errors are, under the strict mark, RT 11/0 and AW 3/8 in the first run and RT 10/1 and AW 4/7 in the re-query; under the flag, RT 8/0 and AW 4/8 in the first run and RT 9/1 and AW 3/7 in the re-query; and under the witness gate (one run), RT 5/1 and AW 3/7. In each of these runs and rules the reversal survives. Later runs outside this protocol are in Appendix S (Table 28); in its retest and instructed arm the flag’s reversal does not recur.

What strictness gives up. In the first run the completeness flag refuses 6/8 of the harmless edits and the witness gate (one run) 5/8, so neither separates a false line nothing uses from a defect that matters. The repair-tolerant witness-plus-ledger rule refuses none of them, so the price belongs to particular rules and not to strictness as such; but that rule’s convention defines a harmless edit as acceptable, so it is not an independent check on this point.

Worked-example candidates. In the first run 3 fatal texts meet the worked example’s selection rule (both graders pass at 5, both mark the text incomplete, both quote inside the edit, a check exists, and the label is undisputed): p6-gpt56-ours-F, shown in Figure 1, deedy-claude-fable-5-final-F and p6-deedy-gpt-5.6-sol-pro-final-F. deedy-claude-fable-5-final-F sits on a third-party base written by Claude Fable 5, the lab of the Claude grader and of the agents that wrote the edits, its check records that the lemma’s conclusion still holds, and its diff would quote third-party lines; p6-deedy-gpt-5.6-sol-pro-final-F re-plants the natural P6 lemma of the public GPT-5.6-Sol default run, and its needed bounded special case is true. p6-gpt56-ours-F is on the author’s own GPT-5.6-Sol base, and its inserted line restates a hypothesis that the proof’s own closing section refutes with the same $A = 15$ example; every grader response on it, in both runs, cites the text’s own statement of the correct set, so by the repair criterion of Appendix L.4 it shares the disputed feature. Its published diff shows 2 hunks from one search/replace edit, and the P6 rubric the witness gate sees names the relevant prime bound, so that example does not illustrate the witness gate.

Table 21: Pass-mark sweep, both graders required, first run and re-query (broken texts accepted of 16, sound texts rejected of 16; post hoc), with the completeness flag for comparison.

Both graders at least	0	1	2	3	4	5	6	7
first run	16, 0	16, 0	16, 0	15, 0	15, 0	13, 0	11, 0	1, 9
re-query	16, 0	16, 0	16, 0	15, 0	14, 0	13, 0	12, 0	1, 8
completeness flag	first run 1, 6; re-query 2, 7							

Table 22: The prespecified per-author report, per grader: broken texts accepted and sound texts rejected on the texts of the six GPT-authored bases and of the other two, first run / re-query (the strict mark and the flag are post hoc).

Grader	Rule	GPT-authored bases		other bases	
		broken acc. (of 12)	sound rej. (of 12)	broken acc. (of 4)	sound rej. (of 4)
GPT-5.6-Sol	pass mark (≥ 5)	11/11	0/0	4/4	0/0
	strict mark ($= 7$)	3/3	6/5	0/0	3/3
	completeness flag	3/3	4/5	1/1	2/2
Claude Opus 5	pass mark (≥ 5)	10/10	0/0	3/3	0/0
	strict mark ($= 7$)	2/2	0/1	0/1	0/0
	completeness flag	2/2	0/1	0/1	0/0
both	pass mark (≥ 5)	10/10	0/0	3/3	0/0
	strict mark ($= 7$)	1/1	6/5	0/0	3/3
	completeness flag	1/1	4/5	0/1	2/2

Table 23: Five rules over the 8 originals and 24 mutants, both graders required, repair-tolerant convention (broken accepted, sound rejected; first run). Name map: score ≥ 5 is the pass mark, score = 7 the strict mark, the completeness verdict the completeness flag, closure rating $S=2$ the closure rule (a written-mechanism rule) and witness $S_{\text{valid}}=2$ the witness rule. The closure and witness rows are reference-guided (their prompts carry the marking scheme; Appendix L.6) and are not information-matched to the rows above; both were run once. As binary gates the pass mark, closure and witness rules are prespecified and the strict mark and flag post hoc; the routing columns are post hoc. In the re-query the strict mark gives (1, 8) and the flag (2, 7); routed, the flag refers 9, passing 2 broken and refusing 1 sound texts.

Rule	as a binary gate		as accept/reject/refer		
	broken accepted	sound rejected	referred	broken accepted	sound rejected
score ≥ 5	13/16	0/16	2/32	13/16	0/16
score = 7	1/16	9/16	12/32	1/16	0/16
completeness verdict	1/16	6/16	10/32	1/16	0/16
closure rating $S=2$	2/16	2/16	9/32	2/16	0/16
witness $S_{\text{valid}}=2$	0/16	5/16	4/32	0/16	1/16

Table 24: Planted-defect gate comparison, every gate and model: false accepts on fatal and gap mutants, false rejects on originals, harmless edits accepted, and locus hits.

Gate	false accepts (F+G, $n=16$)	false rejects (O, $n=8$)	H accepted ($n=8$)	quotes in hunk
ordinary grade, GPT-5.6-Sol	15/16	0/8	8/8	18/23
ordinary grade, Claude Opus 5	13/16	0/8	8/8	24/34
ordinary grade, both models	13/16	0/8	8/8	42/57
score = 7, GPT-5.6-Sol	3/16	1/8	0/8	–
score = 7, Claude Opus 5	2/16	0/8	8/8	–
score = 7, both models	1/16	1/8	0/8	–
completeness verdict, GPT-5.6-Sol	4/16	0/8	2/8	–
completeness verdict, Claude Opus 5	2/16	0/8	8/8	–
completeness verdict, both models	1/16	0/8	2/8	–
closure S, GPT-5.6-Sol	3/16	0/8	6/8	–
closure S, Claude Opus 5	8/16	0/8	8/8	–
closure S, GLM-5.3	8/16	0/8	8/8	–
closure S, GPT and Claude	2/16	0/8	6/8	–
closure S, all three raters	2/16	0/8	6/8	–
witness S_{valid} , GPT-5.6-Sol	0/16	0/8	3/8	20/23
witness S_{valid} , Claude Opus 5	0/16	0/8	7/8	16/18
witness S_{valid} , both models	0/16	0/8	3/8	36/41
witness objections + reference (RT)	2/16	0/8	8/8	–
witness objections + reference (AW)	0/16	0/8	3/8	–
grader errors + reference (RT)	1/16	0/8	7/8	–
grader errors + reference (AW)	0/16	1/8	0/8	–
all objections + reference (RT)	1/16	0/8	7/8	–
all objections + reference (AW)	0/16	1/8	0/8	–

M Paired Critique Pilot: Protocol and Delivery Outcomes

Status of the protocol. The pilot is *protocol-first and exploratory*: its protocol was never registered, and it was issued in the same step as the launch command, not before it. The run records (`meta.json`, UTC) put the launch at about 21:44 on 2026-09-05, the four runs from the $S=2$ seed finishing 21:52–21:58, and the twelve others starting 22:05 after a restart with more parallel workers (the partial first-launch runs were lost). The times in the protocol’s header (22:20) and in its extension note (22:40) were typed from memory and are wrong; the extension note was written at about 22:34, when 14 of 16 runs had finished. No immutable pre-run record exists: the first commit containing the file postdates every primary run. The hypotheses were not changed after any outcome was seen. A dated correction stating all this is appended to both released copies of the protocol (`pilot_critique/PREREG.md`, `pilot_critique_ext/PREREG.md`); the original header is left in place. The four starting texts are hash-identified and their S values are the corpus labels of Appendix F, on which both annotators agree.

Delivery outcomes. Delivery (a write-up returned before the requested token cap), closure rating (S of the returned text), and gate acceptance are separate endpoints, none of them validity; a run that ends inside its reasoning returns no text and is not scored $S=0$. *Primary runs* (requested cap 60k tokens, Kimi K3 via the Moonshot API, tool-free, one call each): 4 of 16 returned text, all four from the $S=2$ seed (`finish_reason=stop`, 9–15 kB each); the 12 continuations of the three $S<2$ texts all ended at the cap inside their reasoning (HTTP 200, `finish_reason=length`, 155k–243k reasoning characters, no output). The reasoning text was not retained, only its length, so nothing about its mathematical content is observed. All four delivered texts were graded and read (Table 25). *Post-hoc extension* (identical prompts; the twelve $S<2$ runs re-run into `pilot_critique_ext/`). With a 200k budget (post-hoc extension), 6 of 12 runs from $S<2$ texts returned a write-up ($S=1$ start: 1/2 with critique, 2/2 without; the two $S=0$ starts: 2/4 and 1/4); of the 6 delivered texts, 4 are rated $S=2$ by both raters and 3 of 5 reviewed are accepted by the gate (1 gate review missing at freeze); the missing gate review is one the provider never returned. Grading of the delivered extension texts (both annotators, the tool-free gate, and a read produced by a model under the author’s direction and reviewed by the author) is otherwise complete; Table 25 reports delivery, closure rating, and gate acceptance per seed and condition. The extension is a budget sensitivity check, not a primary outcome. Larger caps returned more texts; the sample establishes no critique advantage and cannot identify an absence of effect. From the closed seed the model’s own rewrite found both defects, and one no-critique 200k continuation from the $S=0$ Problem-3 text produced a gate-accepted proof: discovery inside one continuation is rare here, not absent.

Table 25: Paired pilot: delivery, closure rating, and gate acceptance per start text and condition (60k runs; post-hoc 200k extension).

Start text	delivered, 60k		delivered, 200k		of delivered: closure $S=2$, gate accepted	
	C	N	C	N	C	N
Kimi P3 seed, $S=2$, local defects	2/2	2/2	–	–	2/2, 2/2	2/2, 2/2
GLM-5.2 P3 candidate, $S=0$	0/2	0/2	1/2	1/2	0/1, 0/1	1/1, 1/1
public Kimi P6 file, $S=1$	0/2	0/2	1/2	2/2	1/1, 1/1	2/2, 1/2 (1 missing at freeze)
Nemotron P6 candidate, $S=0$	0/2	0/2	1/2	0/2	0/1, 0/1	–

C/N = with/without the frozen critique; delivered = write-up returned before the cap; closure rating $S=2$ by both blind annotators; gate = acceptance by a tool-free GLM-5.3 review (model judgments, not validity).

Grading. Outputs were graded under blind identifiers (a hash prefix of the output; neither case nor condition visible to the graders) by GLM-5.3 with thinking and by Kimi K3 under the per-problem progress rubric, with $S=2$ requiring both annotators (the corpus’s conservative rule), and by a tool-free GLM-5.3 review under the reviewer-swap rubric and the fail-closed predicate of Appendix B. Kimi K3 grading its own outputs is a same-model comparison; GLM-5.3 is the cross-lab grader. The protocol, every prompt (redacted only by hashing the draft), every output, and every grade are released.

Protocol. The protocol as issued, with the dated corrections of its header times and of its “author reads” clause, is in the supplementary archive (`pilot_critique/PREREG.md`, `pilot_critique_ext/PREREG.md` and the addenda `pilot_critique/`

ADDENDUM_2026-09-29.md and pilot_critique_ext/ADDENDUM_2026-09-29.md); the reviewed version printed it here.

N Two Trajectories in Full

The two public Kimi K3 lineages that carry most of the trajectory observations, with every retained observation (sessions from the per-round logs [Das, 2026]; S in the defect-corrected layer, shown as written / valid where the two differ; grades by the campaign’s verifier). Gaps between rows are unobserved: the harness records the tracking file only when the model rewrites it.

Session	retained write	min	tokens (k)	S	what changed
Problem 3 (first-pass 5/7, repaired to 7/7)					
round 1	turn 8	115.5	228	0	closed form stated as a conjecture from small- n values; reduction asserted “proved” but not written; Liu’s division given without argument
round 2	—	117.0	225	—	no tracking-file write (exploration; time cap)
round 3	turns 7, 14	108.1	243	0, 0	closed form now stated as the answer; reduction still asserted; lower bound sketched “modulo write-up”; upper bound open
round 4	turn 14	44.9	87	2	fresh session told “the computational phase is over”; reduction proved by induction, tree-component lower bound and bisection
round 5	—	91.9	39	—	upper bound written for all n ; verifier 5/7 with two localized defects
main	final file	13.5	25	2	repair session; no tracking-file write retained
Problem 6 (first-pass 3/7, repaired to 7/7)					
round 1	turn 21	131.3	154	0	characterization and periodicity assembly asserted, not written; finiteness admitted as the unproved crux
round 2	turn 7	69.8	146	2 / 1	both mechanisms written and credited by both annotators (layer 0: “complete, correct”); the finiteness step applies a self-duality claim
round 3	turn 9	91.8	22	1	proved for finite minimal transversals to a Zorn-extracted transversal that may be infinite, the fatal defect in the verifier’s 3/7, so the defect-corrected layer scores finiteness 0
main	turn 5, final	87.7	58	2, 2	honest partial checkpoint: periodicity kept, finiteness marked not proved while being fixed; not a regression in the defect-corrected layer, since finiteness was never validly written
					session prompt carried an explicit critique of the round-3 file; a new compactness argument (every transversal contains a finite one) closes the crux: the lineage’s first valid $S=2$; re-grade 7/7

O Public Trajectory Statistics

Per-model totals over all sessions of the public campaign [Das, 2026] on the three hard problems, computed from its per-round tool-call logs. “cur.md” counts writes of the tracking file; “no-write sessions” counts sessions that ended (time cap, outage) without writing it. Grades are the campaign’s first-pass verifier grades; Kimi’s P3 and P6 were later repaired to 7.

P	Model	sessions	compl. tokens (k)	tool calls	tok/call	cur.md writes	no-write sess.	grade
2	claude-fable-5	1	141	24	5890	2	0	7
2	gpt-5.6-sol-max	1	93	80	1159	16	0	7
2	gpt-5.6-sol-pro	1	254	55	4609	8	0	7
2	gpt-5.6-sol-xhigh	4	113	132	858	11	2	7
2	kimi-k3	1	171	26	6589	1	0	7
2	gpt-5.6-sol	1	18	19	940	4	0	4
2	deepseek-v4-pro	1	296	80	3698	4	0	2
2	grok-4.5	1	127	83	1535	10	0	2
2	muse-spark-1.1	1	183	12	15283	1	0	2
3	claude-fable-5	2	331	25	13241	3	1	7
3	gpt-5.6-sol-xhigh	1	43	19	2277	4	0	7
3	kimi-k3	6	847	118	7175	4	3	5→7
3	gpt-5.6-sol	1	17	16	1036	3	0	2
3	gpt-5.6-sol-pro	1	58	13	4451	3	0	2
3	grok-4.5	1	273	81	3376	6	0	2
3	deepseek-v4-pro	1	326	55	5936	1	0	1
3	gpt-5.6-sol-max	1	1	1	982	1	0	1
3	muse-spark-1.1	1	224	24	9337	1	0	1
6	claude-fable-5	2	122	12	10160	1	1	7
6	gpt-5.6-sol-pro	1	100	16	6263	2	0	7
6	gpt-5.6-sol-xhigh	2	36	17	2121	4	0	7
6	kimi-k3	4	381	79	4824	4	0	3→7
6	muse-spark-1.1	1	204	16	12771	2	0	2
6	deepseek-v4-pro	1	639	56	11414	1	0	1
6	gpt-5.6-sol	1	21	12	1785	2	0	1
6	gpt-5.6-sol-max	1	6	6	1021	2	0	1
6	grok-4.5	1	7	11	641	1	0	1

P Use of AI Assistance and Provenance

Language models are the object of study in this paper: every grade, closure rating, witness, objection and adjudication reported here was produced by a model, and the prompts, transports and stored responses of these experiments are released, except as stated under Availability below. This appendix states what else AI systems did and what the author did, as far as the session records, the git history and the stored files show. No model is an author, and the author is responsible for the entire content.

The planted-defect study. The design of the planted-defect study is the author’s; AI assistants helped draft design options, and an AI coding assistant (Claude Code) built the study under the author’s direction. A coordinating Claude Fable 5.1 agent selected the bases, wrote the specification (PROTOCOL_PLANTINGS.md) and suggested several of the edits; three Claude Fable 5.1 subagents wrote the 24 edits, their dependency arguments, their intended truth labels and the 16 executable checks; the coordinator launched the gate runs. The author decided the design (three edit kinds per proof, each published as a diff with a dependency argument and a check, with the assistant doing the edits), choosing to run it against the assistant’s recommendation, and did not review individual edits before the gates ran (Appendix L.1).

The author’s comparison with Chen’s notes. The author compared the six GPT-5.6-Sol proofs of the author’s campaign with Chen [2026] and found them consistent; this was not a line-by-line or expert check, and no written record of it was kept. On 5 September 2026 a Claude model also compared each of the six proofs with the notes step by step and reported no conflict; drafts before the reviewed version described that model comparison as a human line-by-line read, and the reviewed version presented the author’s comparison as a human check of the proofs (row Q13). The model comparison is in the session log, not in the released artifacts. On 29 September 2026 a second model check (Claude Fable 5.1 agents, each followed by an adversarial verifier) compared the six GPT-5.6-Sol bases of the planted study with the notes (version of 27 September 2026) and found all six consistent, with one minor scope slip in a public base that nothing depends on (Appendix L.2); it is a model check, not a human or expert review, and its record is not in the released archive either.

Other records credited to the author. The reference ledger (reference_adjudication/LEDGER.md and LEDGER_v2.md), the target-match judgments in witness_discovery/recovery.json, the reads of the pilot outputs (pilot_critique/author_read.md and pilot_critique_ext/author_read.md) and the step-by-step comparisons in closure_adjudicated/adjudication_notes.md were produced by a model (Claude, in Claude Code) under the author’s direction and reviewed by the author. The reviewed version described them as the author’s (“written by the author”, “judged by the author”, “author read”, “we compared”); dated addenda next to each record now say the same as this paragraph. No independent mathematician checked any base, label, objection, rating or comparison.

Internal pre-submission reviews. Before submission, between 5 and 7 September 2026, the manuscript went through repeated rounds of internal acceptance review (labelled “revised” and v3 to v17 in the author’s records). Language models ran every round, and no human reviewer took part. Each round was run at the author’s request in the OpenAI Codex desktop application with the model gpt-6-astra; from the third round on, the main Codex agent reviewed each version against the previous one, a separately spawned Codex subagent without the conversation history wrote a fresh review, and further subagents audited the evidence and the supplementary bundle. The same setup also assessed workshop fit and commented on two revision plans, and Codex made consistency edits to the manuscript and to scripts on 5 September 2026; most revisions were made with Claude Code (Claude Fable 5.1, briefly Claude Opus 4.8, later Claude Opus 5), which also ran a few reviewer-role and brainstorming subagents and a literature-survey subagent on Claude Sonnet 5 (the one that suggested the re-query, Appendix L.1). The author requested each round, passed most of the reports on, approved the resulting changes, and read and commented on drafts. In the reviewed version and in the supplement’s dated records (DEVIATIONS.md, the protocols and the ledgers), “the reviews”, “a reviewer”, “the vN review” and “post-vN review” refer to these internal model-run reviews, not to the workshop’s referees, whom this version calls referees; the review reports are not released. After submission, on 11 September 2026, a challenge panel of four Claude Opus 5 subagents run in Claude Code at the author’s request found several of the issues this version corrects (the marking scheme in the witness and closure prompts, the run dependence of the strict-mark and flag comparison, and

smaller repairs); the same Claude agent re-checked those findings against the released artifacts, and no person verified them independently.

This version. The camera-ready was prepared with Claude Code at the author’s direction. A planning panel of Claude Fable 5.1 agents (29 September 2026) planned the corrections from the referees’ reviews and an evidence dossier recomputed from the stored artifacts; at the author’s request OpenAI Codex (gpt-6-astra) wrote an independent camera-ready proposal, which further Claude Fable 5.1 agents reconciled with the panel’s plan item by item, adopting several of its points. AI agents in Claude Code traced the provenance stated in this appendix from the session records and the git history, and Claude Opus 5.5 agents recomputed every number from the stored artifacts, rewrote the text, prepared the supplement and checked the result. No grader or study model was called and no experiment was run for this version. The author approved the plan and answered its questions about provenance and licensing; nothing is uploaded without the author’s review.

Scripts, prose and checks. Beyond these roles, the author used AI coding assistants (Claude Code with Claude Fable 5.1 and Claude Opus 5, and OpenAI Codex; see the paragraph on internal reviews for the other models) to draft and revise the analysis scripts and much of the prose, under the author’s direction and review. What the paper’s accuracy rests on, concretely: every count in the body is a macro that a script writes from the stored artifacts, and a literal check fails the build on any typed numeral in the body other than years, dates, problem and model names, the grading scale’s cut points and formulas; the figures in these appendices are generated in the same way, apart from a few structural counts written in words; a claim guard, extended for this version (the supplement’s CHANGES.md), checks sentence by sentence that each generated figure is printed with its population, its paired outcome, its run and its post-hoc status. Generation keeps a figure consistent with its computation, not with the sentence built around it. Each printed reference was checked against its source record: for the reviewed version by an AI assistant at the author’s request (arXiv records for title and authorship, web sources for existence, title and date, the cited repository commit for its hash, author and date, and the social-media posts for author, date and content), and for the entries added in this version, on 29 September 2026, against their arXiv, ACL Anthology and OpenReview records.

Availability and licence. Code, data and the stored model responses of the experiments reported here, except the IMO 2025 campaign of Appendix A (the archive’s README.md lists what is left out): <https://github.com/bobbercheng/verification-gap>. The supplementary archive re-checks every stored record and re-runs the paper’s analyses offline (its README.md says how); the counts added in this version are read from the same stored responses, which it ships. It redistributes texts of the public IMO 2026 campaign [Das, 2026] only so that the numbers can be recomputed, with no licence granted (the cited repository states none at the cited commit), and it quotes the IMO problem statements and short passages of published solutions without relicensing them. The author’s own material is licensed by its rights holder, Bobber Cheng, under CC BY 4.0. NOTICE.md in the archive says which files are which.

Q Changes Since the Reviewed Version

Dated 29 September 2026. This version corrects the one the workshop’s referees read: the PDF submitted on 7 September 2026 (458,265 bytes, SHA-256 7355ecd6c0395f57bb180b410743a7ea0a2a8639cabba794e24711749d8f4c9f) and its supplementary archive (18,059,371 bytes, SHA-256 c96fab12d47524f4921ef4459af1380a4e0174da1ab7947116c72a60063019c3). Appendix letters A–P are unchanged; Q, R and S are new. Text corrections made on 2 October 2026 carry no identifier and are listed in the archive’s CHANGES.md.

The corrections that change what a reader should conclude. (1) The witness and closure gates were prompted with the problem’s marking scheme, which the grade prompt forbids, so their comparison with the rules read from a grading response is reference-guided against reference-free, and on P1 the scheme meets the planted edits twice (rows Q14, Q15). (2) The comparison of the strict mark with the completeness flag depends on the run: the flag rejects fewer sound texts in the first run, and neither dominates in the re-query; “no pass mark reaches the flag’s operating point” is removed (Q1, Q2). (3) The witness gate’s false rejections have two denominators: none of the 8 originals, the protocol’s endpoint, but 5 of the 16 texts that are sound under the repair-tolerant convention (Q18). (4) Five intended labels, not four, are disputed, and all five are named (Q10). (5) AI agents wrote the

planted edits, their labels and checks; the author’s comparison of the GPT-5.6-Sol proofs with Chen’s notes was not a line-by-line read; and the internal reviews were model-run (Q8, Q13, Q36).

Reviewed abstract (verbatim, as on OpenReview):

Frontier models solved every IMO 2026 problem within hours. Whether those proofs are correct is decided by a grader, so we broke eight accepted proofs on purpose to see what that decision rests on. One frontier grader gave the same score, 6, to every proof we made false and to every proof we merely made ugly, telling the two apart at chance (0.50 against 0.94 for the other) while quoting the edited passage in 7/13 of the fatal-or-gap texts both graders passed. A threshold can therefore pass a text whose own response quotes an objection to it and marks it incomplete, and that response carries a field that does not: over 24 published single-hunk mutants, 16 with an executable check, no pass mark on the 0–7 scale reaches the operating point of the completeness flag beside it, since 5 accepts 13 of 16 broken proofs while 7 accepts 1 but rejects 9 of 16 sound ones. We therefore ask what an operator can establish without a full expert regrade, and audit every retained artifact of our four-model campaigns and a public nine-run one, where two of our own gates accepted a claim the sequence 2,1 refutes. Relabelling only the cosmetic edits reverses which grader is the more accurate on all 3 strict rules, which agree pairwise on 27–29 of 32 texts. Referring only what two graders dispute costs 4/32 under an obligation-level witness rule, which passes 0/16 broken proofs but refuses 1/16 sound ones, against 10/32 and 1/16 for the flag the graders return for free. The response carried the objection; the number did not. We release the corpus, ratings, witnesses, certificates and ledger.

Reviewed TLDR (verbatim):

We broke eight accepted IMO 2026 proofs on purpose: a pass mark accepts texts whose own grading response quotes an objection and marks them incomplete. The response carried the objection; the number did not.

On OpenReview both are replaced by this version’s abstract and the TLDR “AI agents made 24 single edits to 8 model-accepted IMO 2026 proofs. Requiring both graders, 5/7 passed 13 of the 16 fatal-or-gap edits in each of two runs, though a grader quoted each; 7/7 rejects every fatal edit but most harmless ones too.”.

Full list. Every corrected sentence carries an identifier (Q1–Q37, Q41–Q43, QA1–QA3); the row for each identifier gives the reviewed wording, what replaces it and where, and who raised it. The rows are in `corrections.pdf`, in the supplementary archive and at <https://github.com/bobbercheng/verification-gap/blob/main/corrections.pdf>.

Moved or removed. The reviewed body was rebuilt around the planted-defect study; the supporting studies’ details stay in Appendices A–O under their reviewed letters, and the body no longer summarizes the verifier probes (App. H) or the critique pilot (App. M). The reviewed body’s five-rule table is in App. L.7 (Table 23) with a corrected caption. The limitations paragraph of App. E moved to the body. App. I’s survival, evaluator-choice, instability-source and grading-procedure tables and their paragraphs, and App. M’s verbatim protocol, left the PDF; they are unchanged in the supplementary archive. The reviewed App. L sentence listing individual gate decisions is replaced by the named statements of App. L.4 and L.7.

Added. The author block, the acknowledgments and the page-1 footnote; the release address and NOTICE.md; five references on evaluating graders and proofs [Mahdavi et al., 2025a, Doddapaneni et al., 2024, McAleese et al., 2024, Dekoninck et al., 2025, Panickssery et al., 2024]; the worked example and the rules table of the body, generated from the stored files (Figure 1, Table 1); App. L’s subsections on protocol status, bases, checks, disputed labels, prompts and each arm’s information; the printed prompts (App. B); this list (App. Q); the per-mutant table (App. R); and runs made after submission, outside this paper’s protocol (App. S). Every stored record of the reviewed archive is unchanged; corrections to records are dated addenda next to them (`*/ADDENDUM_2026-09-29.md`).

R Per-Mutant Table

Table 26 lists every planted text by base: its class, the marking-scheme obligation the edit targets (the `lower_bound` or `upper_bound` field of `plantings.json`, printed under the problem scheme’s own letter and name), whether its label is disputed, the planted change (read from `plantings.json` and checked against the texts), whether an executable check exists, each grader’s grade, completeness field and in-edit quote in the first run and the re-query, and each rule’s decision with both graders required. The strict-mark and flag decisions are post hoc (Appendix L.1). These are per-text records, not rates; the mechanism labels of the fatal edits (Class column, as the body counts them) were coded post hoc from the same planted changes by the model panel that planned this version (Appendix P), and no ranking of error types is claimed.

Table 26: Every planted text by base. Grades: first run / re-query, with T or F for the completeness field and a superscript q where that grader quoted inside the edited passage (exact anchoring). Decisions, both graders required: first run / re-query, ✓ accept, ✗ reject. Class: F rows carry the post-hoc mechanism label. Obligation: the marking-scheme field the edited step falls under, the two fields the witness gate scores, by the scheme’s letter and name (Appendices B and F): `lower_bound` is (T) termination on P1, (L) the lower bound on P3, (V) sufficiency on P5 and (F) finiteness on P6, and `upper_bound` is (I) invariance, (U) the upper bound, (N) necessity and (P) periodicity; only on P3 are the two fields a lower and an upper bound. Disp.: one of the five disputed labels. Check: the edit has an executable check of the planted falsity; deletions have none.

Class	Obligation	Disp.	Planted change	Check	GPT	Claude	Pass	Strict	Flag
p1-deedy-kimi-k3-final (P1; base by Kimi K3, public campaign)									
O	–	–	original (no edit)	–	6T/6T	7T/7T	✓/✓	✗/✗	✓/✓
F (false lemma)	(T) termination	yes	false lemma: a non-increasing integer above 1 never reaches 1	check	6F ^q /6F ^q	6F ^q /6F ^q	✓/✓	✗/✗	✗/✗
G	(I) invariance	yes	proof of the Euclidean-step Claim deleted	deletion	6T/6T	6F ^q /7T ^q	✓/✓	✗/✗	✗/✓
H	(T) termination	–	inline formula becomes $\ell = m + n$ instead of $\ell = mn$	check	6F ^q /6F ^q	7T ^q /7T ^q	✓/✓	✗/✗	✗/✗
p1-gpt56-5aca5e52-v1 (P1; base by GPT-5.6-Sol, author’s campaign)									
O	–	–	original (no edit)	–	7T/7T	7T/7T	✓/✓	✓/✓	✓/✓
F (false lemma)	(T) termination	yes	false lemma: a non-increasing nonnegative integer never reaches 0	check	6F ^q /6F ^q	6F ^q /6F ^q	✓/✓	✗/✗	✗/✗
G	(I) invariance	yes	Euclidean justification of identity (3) and zero-exponent case deleted	deletion	7T/7T	7T ^q /7T ^q	✓/✓	✓/✓	✓/✓
H	(T) termination	–	display claims $\text{lcm}(m, n) = mn/g^2$	check	6F ^q /6F ^q	7T ^q /7T ^q	✓/✓	✗/✗	✗/✗
deedy-claude-fable-5-final (P3; base by Claude Fable 5, public campaign)									
O	–	–	original (no edit)	–	7T/7T	7T/7T	✓/✓	✓/✓	✓/✓
F (graph inference)	(L) lower bound	–	false claim: $ E < V $ rules out cycles and loops	check	6F ^q /6F ^q	6F ^q /5F ^q	✓/✓	✗/✗	✗/✗
G	(L) lower bound	–	proof of Lemma 4 (tree-component argument) deleted	deletion	5F ^q /6F ^q	4F ^q /4F ^q	✗/✗	✗/✗	✗/✗
H	(U) upper bound	–	display gives a wrong value for $(1 - u)/2$	check	6F ^q /6F ^q	7T/7T	✓/✓	✗/✗	✗/✗
deedy-gpt-5.6-sol-xhigh-final (P3; base by GPT-5.6-Sol, public campaign)									
O	–	–	original (no edit)	–	7T/7T	7T/7T	✓/✓	✓/✓	✓/✓
F (pigeonhole)	(U) upper bound	–	pigeonhole run over nonempty proper subset sums only	check	6F/6F	6F ^q /6F ^q	✓/✓	✗/✗	✗/✗
G	(U) upper bound	–	proof of the cutting fact (5) deleted	deletion	7T/7T	6F ^q /6F ^q	✓/✓	✗/✗	✗/✗
H	(L) lower bound	–	inserted inline sum $1 + \dots + 2^{k-1}$ given as $2^k - 2$ (true: $2^k - 1$)	check	6T ^q /6F ^q	7T/7T	✓/✓	✗/✗	✓/✗
gpt56-ours-accepted (P3; base by GPT-5.6-Sol, author’s campaign)									
O	–	–	original (no edit)	–	7T/7T	7T/7T	✓/✓	✓/✓	✓/✓
F (graph inference)	(L) lower bound	–	false claim: the rank graph has no cycle or loop	check	6F/6F	6F ^q /6F ^q	✓/✓	✗/✗	✗/✗
G	(U) upper bound	–	overlay construction for nonempty B deleted	deletion	6F ^q /6F ^q	6F ^q /6F ^q	✓/✓	✗/✗	✗/✗
H	(L) lower bound	–	interval marks $2^j \delta$ in place of $(2^j - 1)\delta$	check	6F ^q /6F ^q	7T ^q /6F ^q	✓/✓	✗/✗	✗/✗
p5-gpt56-7324ef70-v1 (P5; base by GPT-5.6-Sol, author’s campaign)									
O	–	–	original (no edit)	–	7T/7T	7T/7T	✓/✓	✓/✓	✓/✓
F (sign)	(N) necessity	yes	sign flipped in an expansion: $(b - c)$ in place of $(c - b)$	check	6F ^q /6F ^q	7T ^q /7T ^q	✓/✓	✗/✗	✗/✗
G	(N) necessity	–	derivation of the quadratic estimate (7) deleted	deletion	7T/7T	5F ^q /6F ^q	✓/✓	✗/✗	✗/✗
H	(V) sufficiency	–	sufficiency display gets the middle term $(x + y + 2C)/2$	check	6F ^q /6F ^q	7T ^q /7T	✓/✓	✗/✗	✗/✗
p6-deedy-gpt-5.6-sol-pro-final (P6; base by GPT-5.6-Sol, public campaign)									
O	–	–	original (no edit)	–	7T/7T	7T/7T	✓/✓	✓/✓	✓/✓
F (false lemma)	(F) finiteness	–	false lemma: finitely many minimal finite transversals	check	6F ^q /6F ^q	6F ^q /6F ^q	✓/✓	✗/✗	✗/✗
G	(F) finiteness	–	proof of Lemma 2 (descent step) deleted	deletion	5F ^q /6F ^q	4F ^q /3F ^q	✗/✗	✗/✗	✗/✗
H	(F) finiteness	–	cardinality display’s exponent raised from $ D - 1$ to $ D $	check	6F ^q /6F ^q	7T ^q /7T ^q	✓/✓	✗/✗	✗/✗
p6-gpt56-ours (P6; base by GPT-5.6-Sol, author’s campaign)									
O	–	–	original (no edit)	–	7T/7T	7T/7T	✓/✓	✓/✓	✓/✓
F (definition)	(F) finiteness	–	prime set Q redefined as the primes dividing A	check	6F ^q /6F ^q	5F ^q /6F ^q	✓/✓	✗/✗	✗/✗
G	(F) finiteness	–	proof of the key lemma deleted	deletion	3F ^q /3F ^q	2F ^q /2F	✗/✗	✗/✗	✗/✗
H	(F) finiteness	–	unused worked example lists 22 as the third term	check	6T ^q /7T ^q	7T ^q /7T ^q	✓/✓	✗/✓	✓/✓

S Post-Submission Runs on the Planted Texts

Status. These are runs made by the author after submission (the reviewed version was submitted on 7 September 2026), on the same 32 planted texts and with the same two graders, between 2026-09-08 and 2026-09-21. They are post-submission and outside this paper’s protocol: the texts, their labels and the reviewed results were known before any of these calls, so every figure in this appendix is post hoc, and nothing in the body rests on them. Their records, filtered to the 32 texts and copied byte for byte, are in the supplementary archive (post_submission/, with summary.json and the script that recomputes it). There are 5 runs: a re-run of the witness gate with the same recorded prompt (2026-09-08 to 2026-09-09); a third run of the grade gate with the same recorded prompt (the later run, 2026-09-08); a witness audit with the marking scheme removed (the answer-key-free arm, 2026-09-11); the grade prompt with three sentences of the audit’s instructions added (the instructed arm, 2026-09-21); and a fourth run of the unmodified grade prompt, made in the same session right after the instructed arm (the retest, 2026-09-21). The rules are the paper’s: the pass mark, the strict mark and the completeness flag for the grade runs, and $S_{\text{valid}}=2$ for the witness runs, each with both graders required and per grader; broken means a fatal or gap edit, and sound means an original or a harmless edit, sound only under the repair-tolerant convention.

The grade runs. Across the 4 runs of the unmodified grade prompt (the first run, the re-query, the later run and the retest), the pass mark accepts between 13 and 14 of the 16 broken texts and rejects none of the 16 sound ones, while the flag’s broken acceptances range from 1 to 4. This paper’s pass-mark count stays 13 of 16 in the first run and the re-query; the later run’s 14 comes from separate calls. In (broken accepted, sound rejected), the later run gives 14, 0 for the pass mark, 1, 9 for the strict mark and 1, 6 for the flag, and the retest 13, 0; 3, 8; and 4, 4 (Table 27). In the retest Claude Opus 5’s own flag accepts 11 of the broken texts and rejects 0 sound ones, so the flag is not a fixed comparator across sessions. Under the strict mark, which lab makes fewer errors reverses with the convention in every grade run here; under the flag it reverses in the first run, the re-query and the later run but not in the retest (per-lab errors, GPT-5.6-Sol/Claude Opus 5: repair-tolerant 8/11, as written 8/19) or the instructed arm (10/12 and 4/20; Table 28). In every one of the 5 stored grade runs of the two graders on these texts (320 responses), the flag is nested between the cuts at 6 and 7: 0 responses are graded 7 without it or 5 or below with it.

The instructed arm. The instructed arm adds one line to the grade prompt (“Judge only what is written. Do not use the length of the text, its confidence, or claims of completeness. Do not repair the argument for the author.”) and keeps the 5–6 band, so it is a partial instruction control, confounded with its session, and not the rubric requiring rejection of unresolved essential errors that a referee asked for. In this run the pass mark accepts 14 of the broken texts and rejects 0 of the sound ones, the strict mark gives 1, 9 and the flag 3, 7; the strict mark’s reversal holds in this run and the flag’s does not.

The witness gate, re-run and without the marking scheme. The re-run of the keyed witness gate reproduced every per-grader decision (0 of 64 changed): it accepts 0 of the broken texts and rejects 5 of the sound ones, all harmless edits, so the gate this paper ran once is stable under a repeat of its prompt. The answer-key-free arm is not fully information-matched: the model chose its own obligations, the prompt keeps “Do not repair the argument for the author”, and it was run once and never retested, so its own run-to-run variation is unmeasured. With both graders required it accepts 0 of the 16 broken texts and rejects 6 of the 16 sound ones, all harmless edits. Neither grader alone achieves both zeros: GPT-5.6-Sol accepts 0 broken texts and rejects 6 sound ones, and Claude Opus 5 accepts 1 broken text (the disputed p1-gpt56-5aca5e52-v1-G) and rejects 0 sound ones. Routed, it refers 7 texts, passing 0 broken and refusing 0 sound ones; those zeros come from the split, since the referrals are the 6 harmless edits GPT-5.6-Sol rejects and the 1 gap edit Claude Opus 5 accepts. Relative to the keyed run, the key-free prompt, which also has the model choose its own obligations, changed 3 pair decisions, 3 of them on harmless edits and 0 on broken texts, and 5 per-grader decisions, among them Claude Opus 5’s on the disputed p1-gpt56-5aca5e52-v1-G (reject to accept). As a binary gate its counts equal the flag’s in the first run once that deletion is set aside (0 and 0 of 15 broken texts accepted, 6 and 6 sound texts rejected), but the harmless edits they reject differ by one swap: the flag rejects p1-deedy-kimi-k3-final-H and the key-free arm deedy-gpt-5.6-sol-xhigh-final-H; against the flag in the re-query (2 broken accepted, 7 sound rejected) there is no tie. On both P1 fatal edits, whose false lemma the P1 marking scheme names, both graders of the key-free arm also reject, as do the strict mark and the flag in both runs: the key did not

decide the P1 outcomes, although it remains a comparability defect. The arm was run once on 32 non-independent texts from 8 bases, so its zero is a count on these texts, not a rate, and it implies no advantage over the strict mark or the flag.

Input identity. The recorded system prompt and text of the later run and the retest equal the first run’s on every call (the instructed arm’s system prompt differs by design), but the model input is not byte-identical: the command-line harness changed, and input-token counts differ, in the retest on every call for both graders (Table 30).

Table 27: The three grade rules in each grade run, per grader and with both graders required: (broken texts accepted of 16, sound texts rejected of 16). The first run and the re-query are this paper’s; the other runs were made after submission, outside its protocol.

Run	Date	Grader	pass mark	strict mark	flag
first run (P)	2026-09-06	GPT-5.6-Sol	15, 0	3, 9	4, 6
		Claude Opus 5	13, 0	2, 0	2, 0
		both required	13, 0	1, 9	1, 6
same-day re-query (R)	2026-09-06	GPT-5.6-Sol	15, 0	3, 8	4, 7
		Claude Opus 5	13, 0	3, 1	3, 1
		both required	13, 0	1, 8	2, 7
later run, same prompt	2026-09-08	GPT-5.6-Sol	14, 0	4, 9	4, 6
		Claude Opus 5	14, 0	2, 1	2, 0
		both required	14, 0	1, 9	1, 6
retest, same prompt	2026-09-21	GPT-5.6-Sol	14, 0	4, 8	4, 4
		Claude Opus 5	14, 0	6, 0	11, 0
		both required	13, 0	3, 8	4, 4
instructed arm	2026-09-21	GPT-5.6-Sol	14, 0	2, 9	3, 7
		Claude Opus 5	15, 0	4, 0	12, 0
		both required	14, 0	1, 9	3, 7

Table 28: Per-lab disagreements with the intended truth over the 32 texts (GPT-5.6-Sol/Claude Opus 5) under the repair-tolerant (RT) and as-written (AW) conventions; “reverses”: in each run, the lab with fewer errors under RT has more under AW. Witness rows: the keyed gate of the first run and the answer-key-free arm, both run once.

Run	Date	pass mark		strict mark			flag		
		RT	AW	RT	AW	reverses	RT	AW	reverses
first run (P)	2026-09-06	15/13	23/21	12/2	4/10	yes	10/2	6/10	yes
same-day re-query (R)	2026-09-06	15/13	23/21	11/4	5/10	yes	11/4	5/10	yes
later run, same prompt	2026-09-08	14/14	22/22	13/3	5/9	yes	10/2	6/10	yes
retest, same prompt	2026-09-21	14/14	22/22	12/6	4/14	yes	8/11	8/19	no
instructed arm	2026-09-21	14/15	22/23	11/4	3/12	yes	10/12	4/20	no
keyed witness (P)	2026-09-06			5/1	3/7	yes	(witness rule)		
answer-key-free witness	2026-09-11			6/1	2/9	yes	(witness rule)		

Table 29: The witness-gate arms: as a binary gate (broken texts accepted, sound texts rejected) per grader and with both required, and routed (referred of 32, broken passed, sound refused).

Arm	Date	GPT-5.6-Sol	Claude Opus 5	both required	routed
keyed witness, first run (P)	2026-09-06	0, 5	0, 1	0, 5	4, 0, 1
keyed witness, re-run	2026-09-08/09	0, 5	0, 1	0, 5	4, 0, 1
answer-key-free witness	2026-09-11	0, 6	1, 0	0, 6	7, 0, 0

Table 30: Input identity of the later grade runs against the first run: recorded text and system-prompt hashes, GPT-5.6-Sol input tokens and Claude Opus 5 total input tokens equal to the first run’s (differences in tokens where not).

Run	Date	text hash	system hash	GPT-5.6-Sol input tokens	Claude Opus 5 total input
later run, same prompt	2026-09-08	64 of 64	64 of 64	26 of 32 (others −190 to −13)	32 of 32
retest, same prompt	2026-09-21	64 of 64	64 of 64	0 of 32 (others −1187 to −894)	0 of 32 (others +713 to +4332)
instructed arm	2026-09-21	64 of 64	0 of 64	0 of 32 (others −1155 to −862)	0 of 32 (others +757 to +4376)